

PREPARATORY PROBLEMS



NATIONAL CHEMISTRY CONTEST CUBA 2020



Ministerio de Educación
República de Cuba

PREPARATORY PROBLEMS

National Chemistry Contest

Cuba 2020

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Preface

Dear competitors

We introduce the Preparatory Problems for the National Chemistry Contest for high school for the third time. Mentors and students from the country's different provinces have accepted this material well.

It is worth clarifying that all the current contest programs, past-year exams, and other materials discussed with the mentor teachers during the national seminars are still valid. Naturally, it is impossible to cover all the objectives of the competition program in the same exam. For this reason, the problems that appear here allow the exercise of the *Topics of Advanced Difficulty*, which are the contents that will be evaluated in greater depth in the next edition of the national contest (2020). On this occasion, eleventh and twelfth grades should pay special attention to equilibria involving electrolytes poorly soluble in water and cycloaddition reactions, respectively.

Due to its importance, we consider it opportune to include the updated contents of Organic Chemistry (Annex 3). The new list contains the chemical reactions in the previous booklet, for example, structure, bonding, reactivity and identifying organic substances. The annexes also include the results of the Cuban Olympians in the national competition and the international Olympics this year. Although this summary is not the official 12th-grade program yet, we strongly recommend its use as reference material and study guide.

The problems proposed here and the solution methodology will be discussed in October during the National Seminar for Teacher Trainers at the National Training Center in Havana in the Eduardo García Delgado pedagogical school. Provincial mentors from the National Preselection and invited professors from the University of Havana will participate in this event.

Many successes in the following events!

El author

Havana, October 2019

Index

Constants, conversions and equations.....	5
Periodic table of the elements.....	5
Problems.....	6
Topics of Advanced Difficulties.....	6
Problem 1. Chemical formulas.....	7
Problem 2. Superconductors.....	8
Problem 3. Metals.....	9
Problem 4. Metal plus non-metal equal.....	9
Problem 5. Identification of samples.....	11
Problem 6. Nitrogen.....	11
Problem 7. Multiple questions.....	13
Problem 8. Hydrazine.....	15
Problem 9. Ammonia and phosphine.....	16
Problem 10. Reticular energy.....	18
Problem 11. Dragons.....	18
Problem 12. Hydrogen bromides.....	19
Problem 13. Kinetics of peroxydisulfate ion.....	20
Problem 14. Krioscopy.....	21
Problem 15. Impurities.....	21
Problem 16. Chemical equilibrium.....	22
Problem 17. Hydrogen sulfide.....	23
Problem 18. Dissolution and solubility.....	23
Problem 19. Magnesium determination.....	24
Problem 20. Redox reactions.....	25
Problem 21. Iron(II) hydroxide.....	27
Problem 22. Analysis of an alloy.....	28
Problem 23. Thujopsene.....	28
Problem 24. Warburganal.....	30
Problem 25. Shikimic acid.....	31
Problem 26. Tamiflu.....	32
Problem 27. Organic synthesis.....	34
Answers.....	38
Annex 1. National Chemistry Contest 2019.....	75
Annex 2. International Chemistry Olympiads in 2019.....	79
Annex 3. Organic Chemistry Syllabus for the National Contest.....	81

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases contents	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	The third principle of thermodynamics	$Q = \Delta E + W$
Zeroth in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equations	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																	
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

Periodic table from the theoretical exam of the 51 ICHO.

PROBLEMS

Topics of advanced difficulties

Stoichiometry: Calculation of transformation ratios, the composition of a mixture and empirical formulas of organic and inorganic compounds.

Structure: Atomic structure; periodic table and periodicity. Types of chemical bonds. Representation of molecules and ions and description of the covalent bond using Lewis, TEV, TOM and TRPEV Formulas; resonance theory and hybridization theory.

Descriptive: Fitting of equations; ion-electron method (only for 11 and 12 degrees). Chemical properties of simple and compound substances of the most important representative and transition elements; identification of ions/substances. Hydrogen binary compounds.

Thermochemistry: Enthalpy variation. Hess's law. Formation, combustion and bond enthalpies. Specific heat; calorimetry. Lattice energy; Born Haber cycle.

Kinetics: Reaction speed. Effect of concentration; reaction order and rate law; initial velocity method. Activation energy; Arrhenius equation. Reaction mechanisms; Steady State approximation. Catalysts.

Dissolution: Ways of expressing the concentration and composition of mixtures. Electrolyte strength and solubility rules. Colligative properties of aqueous solutions of nonelectrolytes and electrolytes: vapor pressure, osmotic pressure, freezing point depression, and boiling point elevation.

Chemical equilibrium: Relations between the thermodynamic magnitudes (T , ΔH , ΔS , ΔG) and the equilibrium constant; applications in equilibria in the gas phase, dissolution-precipitation and redox processes. Factors influencing the dissolution-precipitation equilibrium of ionic substances in water: temperature, common ion effect, and pH; calculation of solubility of ionic substances in complex systems.

Electrochemistry: electrode potential. Galvanic cell; electromotive force. Electrolytic cell; decomposition potential; Faraday's laws. Potential diagrams. Relationship between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; Nernst equation.

Organic: Cycloaddition reactions. Mechanisms of the main types of organic reactions. Diels-Alder reaction; hydroxylation of alkenes with OsO_4 ; ozonolysis; Simmons-Smith reaction.

Problem 1. Chemical formulas¹

One method of determining an organic compound's empirical formula is to complete the combustion of a sample and analyze the resulting mixture of products. The result of the combustion of organic substances with dioxygen depends on the conditions in which it is carried out. When excess dioxygen occurs, carbon dioxide and water are formed. Still, other products are obtained if the dioxygen is lacking or atoms of elements other than carbon or hydrogen are present. Two situations in which this method was used are described below.

I- A 4.71 g sample of a solid organic substance **A** was introduced into a previously evacuated 2.00 L container. In the vessel, dioxygen was added until a pressure of 390.4 kPa was reached at 25 °C. After ignition, a solid residue of black color and mass 0.240 g was obtained. In addition, 11.0 g of carbon dioxide and 2.70 g of water were formed, and all the dioxygen was found to have been consumed.

1.1. Find the empirical formula of **A** if it is known to consist of carbon, hydrogen, and oxygen. Neglect the vapor pressure of water and the volume of solids.

II- The complete combustion of a liquid mixture of two alkanes, **B** and **C**, originated 61.6 g of carbon dioxide and 29.7 g of water. **B** is the higher homolog (it has one more carbon atom in its molecular formula) of **C**. It is known that the ratio between the amounts of **B** and **C** substances in the mixture is 3:2, respectively.

1.2. Find the molecular formulas of **B** and **C**.

¹ National Chemistry Contest, Cuba, 2019. First Day. Problem 3.

Problem 2. Superconductors²

Yttrium oxide, barium carbonate and copper oxide react in a solid-state reaction at high temperature (900 °C) to form the superconductor **A** which contains 13.4%Y, 41.2%Ba and 28.6%Cu.

- 2.1. Assuming that the only element unaccounted for in **A** is oxygen, determine the empirical formula of **A**.
- 2.2. Given that the oxidation state of yttrium is +3 and barium is +2, calculate the average oxidation state of copper.
- 2.3. Reduction of **A** on a thermogravimetric analyzer at 200 °C in hydrogen reduces all the copper 3+ in the material to copper 2+ and produces compound **B**. All the other elements remain in the same oxidation state. Given that the starting mass was 84.2 mg, what would be the mass of the remaining compound **B** after reduction?

Problem 3. Metals³

We are studying the following seven metals: aluminum, zinc, silver, calcium, potassium, copper and iron. Answer the following questions by chemical equations.

- 3.1. Which of these metals will “dissolve” in water?
- 3.2. Which of the remaining “dissolves” in hydrochloric acid?
- 3.3. Which of the those left over from 3.1 evolves hydrogen with a solution of sodium hydroxide?
- 3.4. How can you dissolve the metals that have proved to be unreactive previously?

There are a solution of lead(II) ions and a solution of copper (II) ions with equal concentrations. Two plates having the same size and mass of an unknown metal Me, which forms Me^{2+} ions, are put into these solutions, one in each. Copper and lead precipitate on the plates. Later on, the plates are removed simultaneously from the solutions, dried and weighed. During these procedures all the precipitated metal remains completely on the plates. The plate taken out of the lead(II) solution shows an increase

² England Chemistry Olympiad, 2001. Round 1. Problem 3.

³ German Chemistry Olympiad, 2006. Round 3. Problem 3-2.

of mass of 19%, the other one a decrease of mass of 9,8%, relating to the starting mass of the plates. Assume that the reaction rate of the precipitation of the metal is the same in both solutions.

3.5. Assume that the reaction rate of the precipitation of the metal is the same in both solutions. Determine the unknown metal.

Problem 4. Metal plus non-metal equal...⁴

Non-metal **A** exists as two simple substances, **B** and **C**, among which **B** is the most stable. Both substances are gases at room temperature. **C** can be obtained by submitting **B** to an electrical discharge, it has a strong penetrating odor and ochre color. **A** forms hydrogenated compounds **D** and **E**, both liquids. **D** is an excellent solvent at room temperature; while **E** is thermically unstable and decomposes slowly to give **D** and **B**. **E** is oxidant, and it is employed as whitening and as fuel.

Non-metal **F** exists in form of the gaseous simple substance **G** which is very stable. **F** forms binary hydrogenated compounds **H** and **I**. **H** is a gas at room temperature, it has a strong penetrating odor, and it is used as solvent at very low temperatures. Liquid **I** is toxic, and it is employed as fuel for missiles, space rockets and satellites.

compounds	D	E	H	I
% of hydrogen (in mass)	11,2	5,9	17,8	12,6

4.1. Find the identity of no-metal **A** and the formula of compounds **B-E**.

4.2. Find the identity of non-metal **F** and the formula of compounds **G-I**.

Metal **J** forms with non-metal **A** compounds **K**, **L**, **M** and **N**. **K** and **L** are white solids and **M** is a red-greyish solid. **J** reacts with **B** at room temperature to give **K**, but if there is an excess of metal, **L** is obtained instead, because the latter is more stable. **L** is also obtained by heating **K**. The solutions of **K** in **D** are employed as whitening agents due to the formation of **E**. besides, **K** is employed in protection masks, fire extinguishers and in submarines as a source of **B**. **M** can be obtained by reaction between **C** and with the dusted anhydrous hydroxide of **J** at low temperature. This compound is unstable and explodes violently forming **N**. **N** can also be obtained by reaction between metal **J** dissolved in **H** with **B**.

⁴ Training for the chemistry olympiads, Sancti Spíritus, Cuba, 2016. Fragment from the original problem.

Metal **J** forms with non-metal **F** binary compounds **O** and **Q**. When **J** is dissolved in **H** at low temperature, a marine blue solution is formed which in the presence of catalytic amounts of Fe(III) slowly releases an inodorous gas and originates **P**. **O** and **Q** are obtained in an indirect manner.

Compound	K	L	M	N	O	P	Q
% of metal J (in mass)	59,0	74,2	32,4	41,8	35,4	58,9	83,1

4.3. Find the formulas of compounds **K-N** and tell which is metal **J**.

4.4. Find the formulas of compounds **O-Q**.

The reaction of certain derivative **J** with water forms a non-colored and inodorous gas whose combustion emits a characteristic sound known as “dogs bark”. The resultant solution is turned red in the presence of phenolphthalein. 1,21 g of an unknown substance release 1,13 L of gas at 0 °C and 101,3 kPa of pressure.

4.5. Find the formula of the unknown substance.

A mixture of **J** oxide and zinc oxide with mass 7,17 g was reacted with 1,00 mol·dm⁻³ hydrochloric acid and 200 cm³ of the acid solution were consumed.

4.6. Find the composition of the sample in mass percent.

Problem 5. Identification of mixtures⁵

A student was assigned the task of identifying the substances composing two solid mixtures (mixture A and B). It is known that mixture A can contain SiO₂, Na, NiCl₂, CrCl₃, CoCl₂ and Ba(NO₃)₂; while mixture B may be formed by Zn, AgNO₃, Ag, Cu(NO₃)₂, KI and Fe. The student reacted every mixture with common inorganic reagents and found the following results:

Mixture A: It is green. It completely dissolves in water originating a green solution and there isn't any evidence of strong exothermic reaction. Upon addition of sodium hydroxide to the resultant solution, a green precipitate is obtained which is partially soluble in excess of reagent forming a bright green solution. The solid portion is dissolved with hydrochloric acid. Upon addition of sulfuric acid, the original solution does not form any precipitate.

⁵ National Chemistry Contest, Cuba, 2018. First Day. Common Exam. Problem 2.

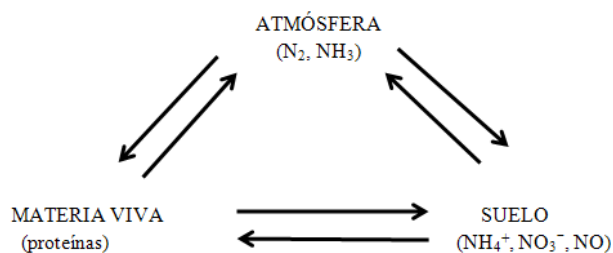
Mixture B: It is grey colored. It dissolves partially in water originating a colorless solution. After allowing to stand several minutes, the solid portion is inert towards hydrochloric acid or sodium hydroxide. On the contrary, when the colorless solution is treated with sodium hydroxide, a white precipitate is formed, which is completely soluble in excess of reagent.

5.1. Tell the substances composing samples A and B.

5.2. Write the equations corresponding to the chemical reactions happening during the identification.

Problem 6. Nitrogen⁶

Nitrogen is a chemical element presented in different forms in nature. The abundance of this element is a third of carbon. It is found free in the atmosphere, in the form of diatomic molecules, constituting 78,03% in volume of it. Nitrogen was recognized as an independent substance in 1772, by the Scottish botanic and chemist Daniel Rutherford (1749-1819), who demonstrated that it was unable to sustain life or combustion. In the living beings, nitrogen constitutes 18% approximately, taking part in the proteins. The path walked by the nitrogen atoms in nature is known as Nitrogen Cycle. This cycle constitutes a very fragile equilibrium, because of that the man must restitute the missing nitrogen in many occasions, via the addition of fertilizers, bacterial control, etc.



There are nitrogen compounds in all the formal oxidation states from -3 up to +5 and nitrogen oxides in the five oxidation states, from +1 up to +5. One of nitrogen compounds is ammonia, very employed as raw material in several synthesis at industrial scale.

6.1. Represent the electronic distribution of N₂ molecule by the Molecular Orbital Theory.

6.2. Explain why it is more difficult to dissociate the molecule of N₂ than the ion N₂⁺.

⁶ X Centroamerican and VIII Caribbean Chemistry Olympiad, Cuba, 2016. Inorganic Chemistry. Problem 1. Adaptation.

- 6.3.** Represent the Lewis' structure of nitrogen oxides at the oxidation states 1+, 3+ and 5+. Name the compounds.
- 6.4.** The nitrogen oxide with oxidation state 5+ exist in an ionic form in the solid state. Represent the electronic structures of the cation and the ion of this oxide according to the Theory of Valence Bond and using hybrid orbitals for the central atom.
- 6.5.** According to the Valence Shell Electron Pair Repulsion model (VSEPR), which is the molecular geometry of NO_2 ? Why? Predict the bond angle.
- 6.6.** In oxidation state 2+, nitrogen exists in the form of two oxides: N_2O_2 at low temperatures and NO at high temperatures. Justify the existence of both compounds taking into account their structures.
- 6.7.** A way to obtain ammonia in the laboratory is by the reaction between a hot alkali and an ammonium salt. Write a chemical equation representing this process.
- 6.8.** Hydrazine (N_2H_4) is a toxic liquid used as fuel. Hydrazine can be synthesized from ammonia and hydrogen peroxide. Write the corresponding chemical equation.
- 6.9.** Phosphorous belongs to the same group than nitrogen in the periodic table. The existence of both phosphorous tri and pentachlorides is known, but there is only nitrogen trichloride. Explain this fact.
- Ammonium nitrate is employed as fertilizer. Nitrate is directly used by plants whereas ammonium is oxidized by microorganisms existing in the soil to nitrite or nitrate and works as a long-lasting enricher.
- 6.10.** Explain why ammonium nitrate is very soluble in water.

Problem 7. Multiple question⁷

Select the right answer in each case.

- 7.1.** When one mole of $\text{Na}_2[\text{Zn}(\text{OH})_4]$ is dissolved in water the result is:

- | | |
|--|--|
| ☐ Two moles of ions | ☐ Three moles of ions |
| ☐ Four moles of ions | ☐ Six moles of ions |
| ☐ Seven moles of ions | |

⁷ X Centroamerican and VIII Caribbean Chemistry Olympiad, 2016. Inorganic Chemistry. Problem 2. Adaptation.

7.2. Samples of 5 g of each of the following substances are given. The one containing the larger number of molecules is:

- | | |
|--|-----------------------------------|
| ☐ Carbon monoxide | ☐ Dioxygen |
| ☐ Methane | ☐ Chlorine |
| ☐ Water | |

7.3. During the Haber process for synthesizing ammonia nitrogen and hydrogen are combined. The right mass ratio N/H for these gases is:

- | | |
|------------------------------|-------------------------------|
| ☐ 1/3 | ☐ 1/7 |
| ☐ 3/1 | ☐ 14/3 |
| ☐ 3/1 | |

7.4. In 120 g of graphite (carbon) there is the same number of atoms as in:

- | | |
|--|--|
| ☐ 10 mol of calcium | ☐ 120 g of sulfur |
| ☐ 100 g of sodium | ☐ 0,5 g of carbon dioxide |
| ☐ 120 mol of hydrogen | |

7.5. Among the following oxides the one with amphoteric behavior is:

- | | |
|--|--|
| ☐ SO_2 | ☐ Al_2O_3 |
| ☐ CaO | ☐ MgO |
| ☐ Na_2O | |

7.6. The analysis of the composition in mass percent of an aluminum and chlorine compound is 20,22 y 79,78, respectively. Therefore, a plausible molecular formula for such compound is:

- | | |
|---|---|
| ☐ AlCl_2 | ☐ Al_2Cl_7 |
| ☐ Al_2Cl_6 | ☐ Al_2Cl_3 |
| ☐ AlCl_3 | |

7.7. In the molecule of SCl_4 the hybridization of sulfur is:

- | | |
|--|--|
| ☐ sp^3 | ☐ sp^3d |
| ☐ sp^2 | ☐ sp^3d_2 |
| ☐ sp | |

7.8. Aspirin or acetylsalicylic acid is one of the more frequently used drugs. It is composed by 60,0% carbon, 4,44 % hydrogen and the rest is oxygen. Its molecular formula is:

- | | |
|---|---|
| ☐ $\text{C}_9\text{H}_2\text{O}$ | ☐ $\text{C}_4\text{H}_4\text{O}$ |
| ☐ $\text{C}_9\text{H}_8\text{O}_4$ | ☐ $\text{C}_7\text{H}_6\text{O}_3$ |
| ☐ $\text{C}_7\text{H}_8\text{O}_4$ | |

7.9. The following chemical equation $\text{HNO}_3(\text{ac}) + \text{CdS}(\text{s}) = \text{Cd}(\text{NO}_3)_2(\text{ac}) + \text{NO}(\text{g}) + \text{H}_2\text{O} + \text{S}(\text{s})$ is balanced with the coefficients from nitric acid to sulfur:

- | | |
|--------------------------------------|--------------------------------------|
| ☐ 4,3,3,4,2,3 | ☐ 8,3,3,2,4,3 |
| ☐ 8,3,3,1,2,3 | ☐ 6,3,6,2,2 |
| ☐ 6,3,3,2,4,3 | |

7.10. 7,275 g of a sample of a mixture containing silver are dissolved in nitric acid. Then, potassium chloride is added and all the silver is stored in 9,00 g of silver chloride. The mass of silver contained in the original sample is:

- | | |
|----------------------------------|----------------------------------|
| ☐ 6,28 g | ☐ 6,78 g |
| ☐ 75,00 g | ☐ 67,74 g |
| ☐ 80,83 g | |

7.11. Among the following metals the one reacting more violently with water is:

- | | |
|-----------------------------|-----------------------------|
| ☐ Cs | ☐ Mg |
| ☐ Na | ☐ K |
| ☐ Ca | |

7.12. From the following pairs of chemical species, the isoelectronic ones are:

- | | |
|---------------------------------------|---|
| ☐ Ne and Ar | ☐ F^- and Cl^- |
| ☐ Ne and F^- | ☐ Na^+ and K^+ |
| ☐ K and K^+ | |

7.13. Among the following statements the right one is:

- ☐ Oxygen and ozone are constituted by atoms of different elements.
- ☐ Chloride anions and potassium cations do not have the same number of electrons.
- ☐ The atoms belonging to elements of IA group have high ionization energy.
- ☐ The fluor atom is larger than the bromo atom.
- ☐ Chlorine atoms have one electron fewer than chlorine ions.

7.14. Iron has atomic number $Z = 26$. In the Fe^{2+} ion in the gaseous state the number of unpaired electrons is:

- | | |
|--|--|
| ☐ There isn't any | ☐ Two electrons |
| ☐ Four electrons | ☐ Six electrons |
| ☐ Eight electrons | |

7.15. It is possible to declare that $^{14}_7N^{3-}$ ion has:

- | | |
|---|---|
| ☐ Seven electrons | ☐ Ten electrons |
| ☐ Fourteen neutrons | ☐ Fourteen protons |
| ☐ The same number of protons and electrons | |

7.16. Element X has atomic number $Z = 23$. Therefore, the electronic configuration for X^{2+} ion is:

- | | |
|---|--|
| ☐ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$ | ☐ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$ |
| ☐ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$ | ☐ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$ |
| ☐ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ | |

7.17. Among the following atoms the one with lowest secondary ionization energy is:

- | | |
|-----------------------------|-----------------------------|
| ☐ Na | ☐ O |
| ☐ K | ☐ Ne |
| ☐ Ca | |

7.18. Among the listed species the one with smallest volume is:

- | | |
|------------------------------------|---------------------------------|
| ☐ O^{2-} | ☐ Na^+ |
| ☐ Mg^{2+} | ☐ Ne |
| ☐ Na | |

7.19. Which of the following chemical species does not exist?

- | | |
|-----------------------------------|----------------------------------|
| ☐ $BeCl_2$ | ☐ BBr_3 |
| ☐ PCl_3 | ☐ PCl_5 |
| ☐ CO_3 | |

7.20. The geometry of the PCl_5 molecule is:

- | | |
|--|---|
| ☐ tetrahedral | ☐ octahedral |
| ☐ trigonal flat | ☐ trigonal bipyramidal |
| ☐ lineal | |

Problem 8. Hydrazine⁸

Hydrazine, N_2H_4 , is a fuel used in the thrusters of space crafts. It is very popular because doesn't release carbon dioxide. The fast production of hot gasses during the combustion is the source of the rocket trust.

8.1. Write the equation of the decomposition reaction of hydrazine in ammonia and nitrogen, which is a collateral reaction happening in the process.

Hydrazine can be obtained by a reaction between ammonia and hydrogen peroxide.

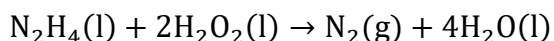


8.2. When hydrazine passes through and adequate catalyst, it decomposes in the parent elements. Calculate the standard enthalpy variation for this process.

ΔH_f° ($\text{kJ} \cdot \text{mol}^{-1}$): $NH_3 = -46,1$; $H_2O_2 = -187,8$; $H_2O = -285,8$.

The first fighter plane thrustured by a rocket, Messerschmitt Me 163, was propelled by the reaction of a mixture containing hydrazine and methanol known as "C-Stoff" and hydrogen peroxide ("T-Stoff").

Hydrogen peroxide reacts with hydrazine according to the equation



⁸ National Chemistry Contest, Cuba, 2019. Second Day. 10th grade. Problem 6.

- 8.3.** What is the oxidation number of nitrogen and oxygen in reactants and products.
- 8.4.** Hydrogen peroxide oxidizes methanol to carbon dioxide and water. Write the chemical equation corresponding to this reaction.
- 8.5.** A fighter plane carries 225 L of hydrazine and 862 L of methanol. Use the following data to calculate the heat released under standard conditions when all the fuel is burnt.

$\Delta H_{\text{combustion}}^{\circ}$ ($\text{kJ} \cdot \text{mol}^{-1}$): $\text{N}_2\text{H}_4 = -622,2$; $\text{CH}_3\text{OH} = -726,0$;

Density ($\text{g} \cdot \text{mL}^{-1}$): $\text{N}_2\text{H}_4 = 1,021$; $\text{CH}_3\text{OH} = 0,7918$.

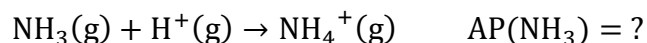
Besides obtention, the storage and transport of fuels possess several challenges. In this context, two central concepts are volumetric and gravimetric energy densities. The volumetric energy density is the recoverable energy divided by the total volume. The gravimetric energy density is the recoverable energy divided by the total mass.

- 8.6.** Calculate the volumetric and gravimetric energy densities for hydrazine and methanol.
- 8.7.** Considering the previous result, choose which of the two substances is more adequate to be employed as fuel.

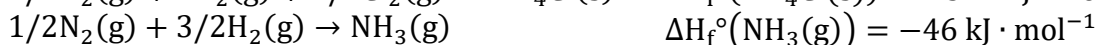
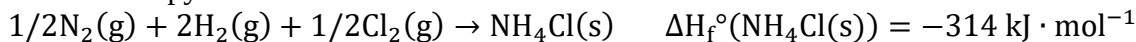
Problem 9. Ammonia and phosphine⁹

Proton affinity (AP) is the energy associated with the process in which a chemical species gains a proton, an H^+ ion. The value of this physical quantity provides quantitative information about the acid-base characteristics.

- 9.1.** Use the data given to calculate the proton affinity of ammonia (NH_3).



Standard enthalpy of formation:



Standard enthalpy of dissociation:

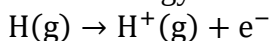


Electron affinity:



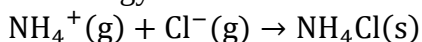
⁹ National Chemistry Contest, Cuba, 2018. Second Day. 10th grade. Problem 5.

Ionization energy:



$$I(\text{H}) = 1313 \text{ kJ} \cdot \text{mol}^{-1}$$

Lattice energy:



$$U(\text{NH}_4\text{Cl}) = -651 \text{ kJ} \cdot \text{mol}^{-1}$$

9.2. A substance is a good base when it can easily accept a proton (ion H^+). Proton affinities of phosphine (PH_3) and ammonia are $-785 \text{ kJ} \cdot \text{mol}^{-1}$ and $-854 \text{ kJ} \cdot \text{mol}^{-1}$, respectively. Which of the two is the best base?

9.3. Phosphine and ammonia molecules are isoelectronic, but HPH and HNH bond angles have an amplitude of $93,5^\circ$ and $107,8^\circ$, respectively. Which atomic orbitals are involved to a more significant extent in the formation of P-H and N-H bonds?

9.4. Represent the electronic configuration of ammonia and phosphine molecules according to the Valence Bond Theory. Only use the orbitals in the valence shell for your analysis.

The enthalpy of a covalent bond A-B, where A and B are two atoms, is the average energy needed to break one mole of these bonds at constant pressure. In diatomic molecules, the bond enthalpy is numerically equal to the dissociation enthalpy.

9.5. The standard mean bond enthalpies for P-H and N-H bonds are 322 and $388 \text{ kJ} \cdot \text{mol}^{-1}$, respectively. Justify why the N-H bond is stronger.

9.6. The strongest chemical bond known to date is that of the dinitrogen molecule. Explain this experimental fact.

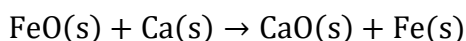
9.7. Calculate the standard enthalpy of dissociation of the dinitrogen molecule. Elaborate on an appropriate thermodynamic cycle and use the data from the problem.

Problem 10. Lattice enthalpy¹⁰

10.1. Using the data, calculate the lattice enthalpies/ $\text{kJ} \cdot \text{mol}^{-1}$ for iron(II) oxide and calcium oxide.

10.2. Which metal oxide has the stronger lattice?

10.3. Calculate the enthalpy change for the following reaction



10.4. Explain whether it would be viable to use this reaction on an industrial scale?

¹⁰ England Chemistry Olympiad, 2001. Round One. Problem 2.

Iron(II) oxide can be readily oxidized to iron(III) oxide by warming in oxygen. Calcium(III) oxide has never been formed.

10.5. Explain why the two metal oxides, iron(II) oxide and calcium (II) oxide, differ.

Standard enthalpy of formation CaO	$-635 \text{ kJ}\cdot\text{mol}^{-1}$
Standard enthalpy of formation FeO	$-278 \text{ kJ}\cdot\text{mol}^{-1}$
Standard enthalpy of atomization Ca	$+178 \text{ kJ}\cdot\text{mol}^{-1}$
Standard enthalpy of atomization Fe	$+416 \text{ kJ}\cdot\text{mol}^{-1}$
Standard molar 1st ionization enthalpy Ca	$+590 \text{ kJ}\cdot\text{mol}^{-1}$
Standard molar 2nd ionization enthalpy Ca	$+1145 \text{ kJ}\cdot\text{mol}^{-1}$
Standard molar 1st ionization enthalpy Fe	$+759 \text{ kJ}\cdot\text{mol}^{-1}$
Standard molar 2nd ionization enthalpy Fe	$+1561 \text{ kJ}\cdot\text{mol}^{-1}$
Standard enthalpy of atomization (O) of O atoms	$+249 \text{ kJ}\cdot\text{mol}^{-1}$
Combined standard molar 1st and 2nd electron affinity (O) of O atoms	$+657 \text{ kJ}\cdot\text{mol}^{-1}$

Problem 11. Dragons¹¹

Dragons are able to breathe fire because the parasitic bacteria that live in their intestines and lungs produce flammable gases. If a dragon exhales sharply, the gas can ignite due to the friction against the rough walls of the larynx. There are several different species of dragons, each of which plays host to a different species of parasitic bacteria. The gas exhaled by the Black Dragon contains 25 mole % of hydrogen, the gas exhaled by the Red Dragon contains 30 mole % of methane (CH_4), and the gas exhaled by the Gold Dragon contains 20 mole % of hydrogen sulfide (H_2S). The gas exhaled by all species of dragon contains 15 mol % of oxygen. The average volume of a dragon's lungs is 5.1 m^3 and, due to the high pressures experienced when a dragon exhales, the molar volume of the gas contained in their lungs is $15 \text{ dm}^3\cdot\text{mol}^{-1}$.

11.1. Calculate the total amount of gas in moles contained in a dragon's lungs.

11.2. Calculate the amount of oxygen in moles contained in a dragon's lungs.

11.3. Write equations for reactions that describe the burning of gas exhaled by the Black, Red, and Gold Dragons respectively.

11.4. How many additional moles of oxygen are required for the complete combustion of one exhalation of the Black, Red, and Gold Dragons respectively?

¹¹ England Chemistry Olympiad, 2000. Round One. Problem 3.

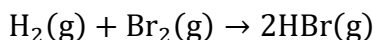
11.5. Calculate the energy released during the combustion of one exhalation of the Black, Red, and Gold Dragons respectively.

Enthalpies of combustion:

$$\Delta H_{\text{comb}}^{\circ}(\text{H}_2) = -240 \text{ kJ}\cdot\text{mol}^{-1}, \Delta H_{\text{comb}}^{\circ}(\text{CH}_4) = -800 \text{ kJ}\cdot\text{mol}^{-1}, \Delta H_{\text{comb}}^{\circ}(\text{H}_2\text{S}) = -520 \text{ kJ}\cdot\text{mol}^{-1}$$

Problem 12. Hydrogen bromide¹²

From a kinetic point of view, many chemical reactions can be classified as chain reactions. In chain reactions, an intermediate formed in one step generates another intermediate in the following step and so on. These intermediates are named propagators. An example of such a process is the reaction between hydrogen and bromine.



The following mechanism has been proposed to explain the results of the kinetics studies.

Initiation	$\text{Br}_2 \rightarrow 2\text{Br}$	$v = k_1[\text{Br}_2]$
Propagation	$\text{Br} + \text{H}_2 \rightarrow \text{HBr} + \text{H}$	$v = k_2[\text{Br}][\text{H}_2]$
	$\text{H} + \text{Br}_2 \rightarrow \text{HBr} + \text{Br}$	$v = k_3[\text{H}][\text{Br}_2]$
Retardation	$\text{H} + \text{HBr} \rightarrow \text{H}_2 + \text{Br}$	$v = k_4[\text{H}][\text{HBr}]$
Termination	$2\text{Br} \rightarrow \text{Br}_2$	$v = k_5[\text{Br}]^2$

12.1. Identify reactants, products and intermediates. State the molecularity of the last step of the reaction. Which is the rate-limiting step?

12.2. Write down mathematical expressions relating the variation of the concentration of the intermediates in time as a function of the rate constants of individual steps and the concentrations of the species involved.

The experimental rate law for the previous reaction is a complex expression (see below); however, under certain experimental conditions, the expression is simplified and a pseudo-order behavior is observed.

$$\text{Rate law: } \frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]}$$

12.3. Which are the partial reaction orders with respect to Br_2 at the exact moment the reaction starts and when it is close to its end? Justify your answer.

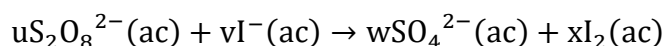
¹² National Chemistry Contest, Cuba, 2018. First Day. Common Exam. Problem 3.

12.4. Find mathematical expressions for the total activation energy and pre-exponential factor, assuming that the reaction rate is $k = 2k_2(k_1/k_5)^{1/2}$.

Problem 13. Kinetics of the peroxodisulphate ion¹³

The peroxodisulphate ion is one of the strongest oxidants that are known, although the oxidation reaction is relatively slow. Peroxodisulphate ions are able to oxidize all halides, except fluoride, to halogens.

The initial rate (r_0) of the iodine-formation as function of the initial concentrations (c_0) of the reactants at 25 °C according to



	$c_0(\text{S}_2\text{O}_8^{2-}) [\text{mol}\cdot\text{L}^{-1}]$	$c_0(\text{I}^-) [\text{mol}\cdot\text{L}^{-1}]$	$r_0 [10^{-8} \text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}]$
(1)	0,00010	0,010	1,10
(2)	0,00014	0,010	1,54
(3)	0,00018	0,015	2,97

13.1. Draw the line-bond structure of the peroxodisulphate ion and determine the oxidation states of all atoms and the integers u, v, w and x.

13.2. Write down the rate equation for the reaction shown above. Determine the order of the reaction and the rate constant.

The activation energy of the reaction mentioned above is 42 kJ·mol⁻¹.

13.3. What temperature (in °C) has to be chosen to decuple the rate constant?

Problem 14. Cryoscopy¹⁴

Chemist often need to prepare a cooling bath to carry out a process at a temperature below the freezing point of water (0 °C) and well above the sublimation point of carbon in acetone (−78 °C). In such cases they mix ice and NaCl. Depending on the amounts employed, temperatures as low as −20 °C can be reached.

¹³ German Chemistry Olympiad, 2012. Round 3. Problem 3-14. Fragment.

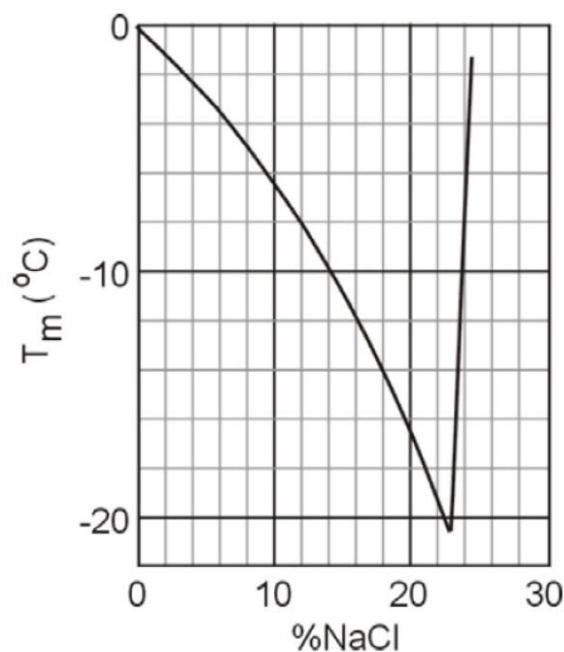
¹⁴ International Chemistry Olympiad, Greece, 2003. Theoretical Exam. Problem 28. Adaptation.

A freezing bath was prepared by mixing 1 kg of ice at 0 °C with 150 g of NaCl in a thermally isolated container.

14.1. State whether the following statements are right or wrong.

- i) The process of mixing is spontaneous.
- ii) The entropy variation during the mixing process is negative.

The figure shows the melting point (T_m) of aqueous solutions of NaCl as function of the solution composition (in mass percent).



14.2. Using the figure, find the melting point of the bath with the composition described.

14.3. If an equal amount of mass of $MgCl_2$ instead of NaCl is employed, what should happen to the melting point?

Problem 15. Impurities¹⁵

During a custom control a parcel with heroin ($C_{21}H_{23}O_5N$), was found, however admixed with lactose ($C_{12}H_{22}O_{11}$). 1,00 g of this mixture was dissolved in water, the solution filled up to 100 mL. This solution had an osmotic pressure of 718 hPa at 25°C.

15.1. Calculate the composition (mass percentage) of the sample.

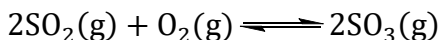
Desiccated anhydrous calcium chloride was stored in an improperly closed vessel. Thus, it was partially hydrated again. A 150 g sample of this material was completely dissolved in 80 g of hot water, then the solution was cooled down to 20°C. On cooling 74.9 g of $CaCl_2 \cdot 6 H_2O$ precipitated. The solubility of calcium chloride in water at 20 °C is 74,5 g of $CaCl_2$ /100 g of water.

15.2. Determine the water content of calcium chloride in the 150 g sample (mole of water per 1 mol of $CaCl_2$).

¹⁵ German Chemistry Olympiad, 2006. Round 3. Problem 3-3.

Problem 16. Chemical equilibrium¹⁶

A mixture of equal volumes of SO₂ and O₂, measured at the same temperature and pressure, has been placed in a reaction vessel of 4,00 L capacity. When the temperature has reached 727 °C the following reaction proceeds:



At the moment of reaching equilibrium the concentrations of SO₂ and SO₃ were exactly the same and the total pressure of the system was 114,8 atm.

16.1. Determine whether the amount of oxygen that was initially in the mixture is sufficient for the combustion with hydrogen releases 1150,20 kJ of energy. Assume that the heat of hydrogen combustion is -241,59 kJ·mol⁻¹.

16.2. Suppose that at 727 °C there is a relation between the concentrations of $Q_c = 3,32 \cdot 10^{-2}$. What happens to the system?

16.3. It is known that at 727 °C the previous gas system has $K_c = 1,29$. Compare this value with that found previously by you. To what conclusion can you arrive?

Another compound containing sulfur, S₄N₄, in its reaction with sodium hydroxide originates the ions thiosulfate S₂O₃²⁻ and trithionate S₃O₆²⁻. In this reaction ammonia is also formed.

16.4. Represent the structure of S₄N₄. It is known that this compound has shape of a symmetric ring, in different atoms alternate, the nitrogen atoms are contained in the same plane and opposed the sulfur atoms interact with Van der Waals forces.

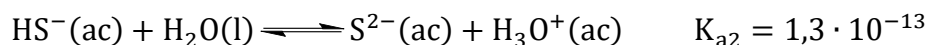
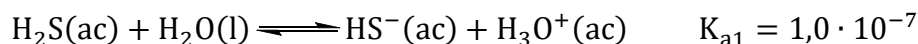
16.5. Write the equation of the described chemical reaction.

16.6. Draw Lewis' structures for S₂O₃²⁻ and S₃O₆²⁻ ions.

¹⁶ National Chemistry Contest, Cuba, 2007. Second Day. 11th grade. Problem 4. Adaptation.

Problem 17. Hydrogen sulfide¹⁷

When H_2S is dissolved in water the solution is saturated at a concentration close to 0.1 mol/L. Three species containing sulfur (H_2S , HS^- , S^{2-}) exist in this solution, and their ratio depends on the acidity of the solution. The equilibrium constants are as follows:



17.1. Calculate the concentration of sulfide ion in a saturated 0.100 M solution of H_2S where the pH is adjusted at 2 by HCl.

A solution contains the cations Mn^{2+} , Co^{2+} , Ag^+ at an initial concentration of 0.010 mol/L each. The following solubility products are given:

$$K_{ps}(\text{MnS}) = 2,5 \cdot 10^{-10}, K_{ps}(\text{CoS}) = 4,0 \cdot 10^{-21}, K_{ps}(\text{Ag}_2\text{S}) = 6,3 \cdot 10^{-50}$$

17.2. Which of these ions will precipitate, if the solution is saturated with H_2S and the pH adjusted to pH = 2. Explain your answer.

The following solubility products are given: $K_{ps}(\text{PbSO}_4) = 1,6 \cdot 10^{-8}$, $K_{ps}(\text{PbS}) = 2,5 \cdot 10^{-27}$.

17.3. How many grams of lead(II) sulfide will precipitate from 1.00 L of a saturated solution of lead(II)-sulfate, if the concentration of sulfide ions is adjusted to be $1,00 \cdot 10^{-17}$ M?

Problem 18. Dissolution and solubility¹⁸

Calcium carbonate and calcium oxalate have the solubility products $K_{ps}(\text{CaCO}_3) = 1,2 \cdot 10^{-8}$ y $K_{ps}(\text{CaC}_2\text{O}_4) = 2,0 \cdot 10^{-9}$ respectively.

18.1. Calculate the concentration of calcium ions in a saturated solution of CaCO_3 .

18.2. Calculate the concentration of calcium ions in a solution saturated with both CaCO_3 and CaC_2O_4 .

In a thermally insulated vessel 25 mL of an aqueous solution of potassium hydroxide are placed. 5 mL portions of nitric acid ($c = 2,00$ mol/L) were added successively. After each addition the temperature was measured:

¹⁷ German Chemistry Olympiad, 2003. Round 3. Problem 3-2.

¹⁸ German Chemistry Olympiad, 2003. Round 4. Problem 4-5.

mL nitric acid	0	5	10	15	20	25	30	35	40	45
temperature in °C	18,0	20,5	22,6	24,4	26,2	27,7	29,0	29,5	29,1	28,7

18.3. What was the mass in the initial sample?

18.4. Explain why there is no linear increase in temperature.

Problem 19. Magnesium determination¹⁹

To determine the amount of magnesium in a solution, a sample of the liquid was first acidified with HCl, then made slightly alkaline by addition of NH_3 and then combined with an excess $(\text{NH}_4)_2\text{HPO}_4$ in water. The precipitate of MgNH_4PO_4 formed was filtered off, washed with diluted aqueous NH_3 , annealed at 1000 °C to constant mass and weighed. Answer the following questions using numerical data given in the end of the text whenever necessary.

19.1. Write down the net ionic equation for the precipitation reaction taking place in course of the analysis.

19.2. Write down the equation for the reaction taking place in the course of annealing if magnesium pyrophosphate, $\text{Mg}_2\text{P}_2\text{O}_7$ is formed.

19.3. When determining the content of magnesium in a granulated medicine preparation calmagin 0.1532 g of the annealed precipitate were obtained from a 1.8005 g sample of calmagin. Calculate the mass percent of MgO in the preparation.

During the precipitation of MgNH_4PO_4 some impurities may coprecipitate such as MgHPO_4 , $\text{Mg}(\text{NH}_4)_4(\text{PO}_4)_2$, $\text{Mg}_3(\text{PO}_4)_2$, $\text{Mg}(\text{OH})_2$, $(\text{NH}_4)_2\text{HPO}_4$ and NH_4Cl . Some of these substances can undergo thermal decomposition at annealing. During the annealing, these substances form $\text{Mg}_2\text{P}_2\text{O}_7$, $\text{Mg}(\text{PO}_3)_2$, unchanged, MgO, HPO_3 y NH_3 , respectively.

19.4. Write down the equations of the corresponding reactions.

19.5. Indicate if the presence of the impurities listed in Table below can lead to an error in the magnesium content as determined by the method described above.

¹⁹ 39th International Chemistry Olympiad, Russia, 2007. Preparatory Problems. Problem 14. Adaptation.

19.6. At what maximum pH value the precipitation of MgNH_4PO_4 may be carried out to avoid simultaneous precipitation of $\text{Mg}(\text{OH})_2$? Assume that the volume of the original sample was 200 mL and the content of magnesium in it was 0.10 g.

To determine the solubility product (K_{sp}) of MgNH_4PO_4 a NaOH solution was added dropwise until the beginning of precipitation to a 100 mL of solution containing 0.010 M MgCl_2 , NH_4Cl and NaH_2PO_4 each. The precipitation started at pH 6.48.

19.7. Calculate K_{sp} . Neglect the volume change during the experiment.

19.8. Draw Lewis' structures for ions PO_4^{3-} , PO_3^- and $\text{P}_2\text{O}_7^{4-}$.

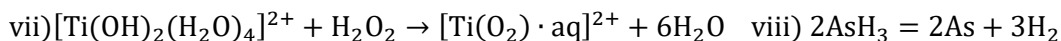
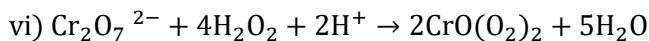
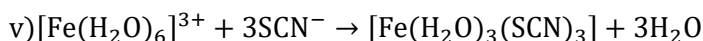
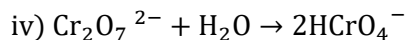
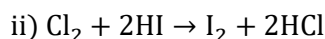
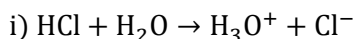
$K_{\text{a}1}(\text{H}_3\text{PO}_4) = 7,1 \cdot 10^{-3}$, $K_{\text{a}2}(\text{H}_3\text{PO}_4) = 6,2 \cdot 10^{-8}$, $K_{\text{a}3}(\text{H}_3\text{PO}_4) = 5,0 \cdot 10^{-13}$; $K_{\text{b}}(\text{NH}_3) = 1,8 \cdot 10^{-5}$;
 $K_{\text{ps}}(\text{Mg}(\text{OH})_2) = 6,0 \cdot 10^{-10}$; $K_{\text{w}}(\text{H}_2\text{O}) = 1,0 \cdot 10^{-14}$

Problem 20. Redox reactions²⁰

The redox reaction is an important type of chemical reactions.

20.1. Explain the concept of "redox reactions" shortly. In doing so use the reaction of sodium with water as an example. Write down the relevant reaction equation(s) and apply oxidation numbers.

20.2. Which of the following equations represent a redox reaction?



Batteries are galvanic elements, electrochemical cells in which redox reactions take place.

20.3. Write down the main difference between primary (common galvanic cells, non-rechargeable) and secondary (accumulators or batteries, rechargeable) galvanic elements.

To harness the energy of a chemical reaction so as to produce an electric current has a long tradition. Small vessels made of clay were excavated from a settlement near to Bagdad (dated 250 BC to 225

²⁰ German Chemistry Olympiad, 2014. Round 1. Problem 1-2.

AD). They contained a copper cylinder and an iron rod. Until today it's not clear whether these vessels point to the use of galvanic elements in those times.

20.4. Which metal could have served as anode, which as cathode if these vessels had been used to provide electrical power? Write down the chemical half equations (anode and cathode). ($E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0,34 \text{ V}$; $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0,41 \text{ V}$)

20.5. Name two more preconditions for the construction of an electrical power providing battery (besides the existence of two electrodes).

20.6. What is the cell potential of the Bagdad battery under standard conditions?

The table on the next page presents some standard half-cell potentials in aqueous solutions at 298 K.

Oxidized form	Reduced form	E°/V	Oxidized form	Reduced form	E°/V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Zn}$	$-0,76$	$\text{I}_2 + 2\text{e}^-$	$\rightleftharpoons 2\text{I}^-$	$+0,54$
$\text{Fe}^{3+} + 3\text{e}^-$	$\rightleftharpoons \text{Fe}$	$-0,04$	$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons \text{Fe}^{2+}$	$+0,77$
$2\text{H}^+ + \text{e}^-$	$\rightleftharpoons \text{H}_2$	$-0,00$	$\text{Cl}_2 + 2\text{e}^-$	$\rightleftharpoons 2\text{Cl}^-$	$+1,36$
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons \text{Cu}$	$+0,34$			

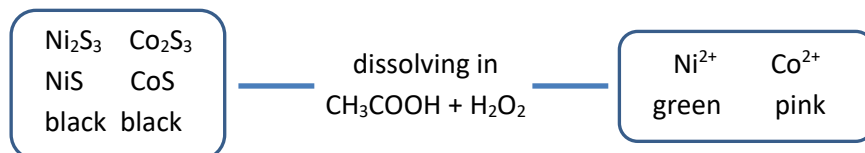
20.7. Which statements concerning the respective experiments are correct? Use the data from the table to find your answer.

i) Granulates of zinc are given into diluted hydrochloric acid:	ii) Splinters of copper are given into diluted hydrochloric acid:	iii) A solution of iron(III) ions is mixed with a solution of potassium iodide:
1. Chlorine evolves.	1. Chlorine evolves.	1. Iodine forms.
2. Hydrogen evolves.	2. Hydrogen evolves.	2. Iron precipitates.
3. Nothing happens.	3. Nothing happens.	3. Nothing happens.
4. Zinc dissolves	4. Copper dissolves.	4. The solution turns blue.

In the separation scheme of cations cobalt and nickel are precipitated as sulfides. For further identification the sulfides are dissolved in $\text{H}_3\text{CCOOH}/\text{H}_2\text{O}_2$ or conc. HNO_3 .

20.8. Is the dissolving of NiS and CoS a redox reaction? Account for your answer!

20.9. Why is it better to use $\text{H}_3\text{CCOOH}/\text{H}_2\text{O}_2$ instead of conc. HNO_3 ?



Problem 21. Iron(II) hydroxyde²¹

The solubility product K_L of iron(II)-hydroxide is to be determined by potentiometry. A galvanic cell is set up. One half cell consists of a slightly acid 0.01 molar Fe^{2+} -solution in which an iron rod is placed. A copper half-cell with a 0,5-molar Cu^{2+} -solution is used as a reference half-cell. A voltage of 0.830V can be read.

Then, the pH-value in the iron half-cell is made exactly 12 with caustic soda lye. But watch out! No air must get into the half cell. A white precipitate can be observed in the solution. The voltage changes to 1.090 V.

The experiment is carried out under standard conditions. Standard redox potential: Cu/Cu^{2+} : +0.340V

21.1. Write down the reaction equations for the processes in the reduction- as well as the oxidation half cells and the total equation.

21.2. Calculate the standard redox potential of Fe/Fe^{2+} .

21.3. What will happen if caustic sodium lye is added to the iron half-cell? Write down the reaction equation.

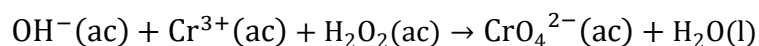
21.4. What would happen if caustic sodium lye was added to an open iron half-cell? Write down the reaction equation.

21.5. Calculate the solubility product K_L of iron(II)-hydroxide. Write down as well the $\text{p}K_L$ -value of $\text{Fe}(\text{OH})_2$.

Problem 22. Analysis of an alloy²²

A sample of 1500 mg of an alloy containing silver, copper and chromium was dissolved and the resultant solution containing ions Ag^+ , Cu^{2+} and Cr^{3+} was diluted exactly to 500 cm^3 . A tenth of the volume of the solution was taken for later procedures.

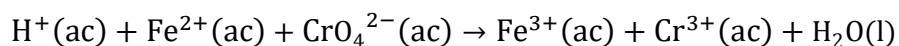
After eliminating silver and copper, chromium was oxidized according to the non-balanced equation:



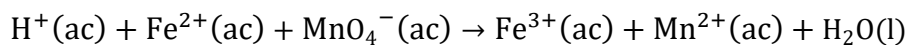
²¹ German Chemistry Olympiad, 2004. Round 3. Problem 3-17.

²² 7th International Chemistry Olympiad, Hungary, 1975. Theoretical Exam. Problem 3.

Then 25,00 cm³ of the 0,100 M solution of an Fe(II) salt were added. The next reaction (written in a non-balanced form) took place:



For the oxidation of unoxidized Fe(II) salt in the solution, 17,20 cm³ of a 0,020 M KMnO₄ solution were added. The reaction (non-balanced) happening is:

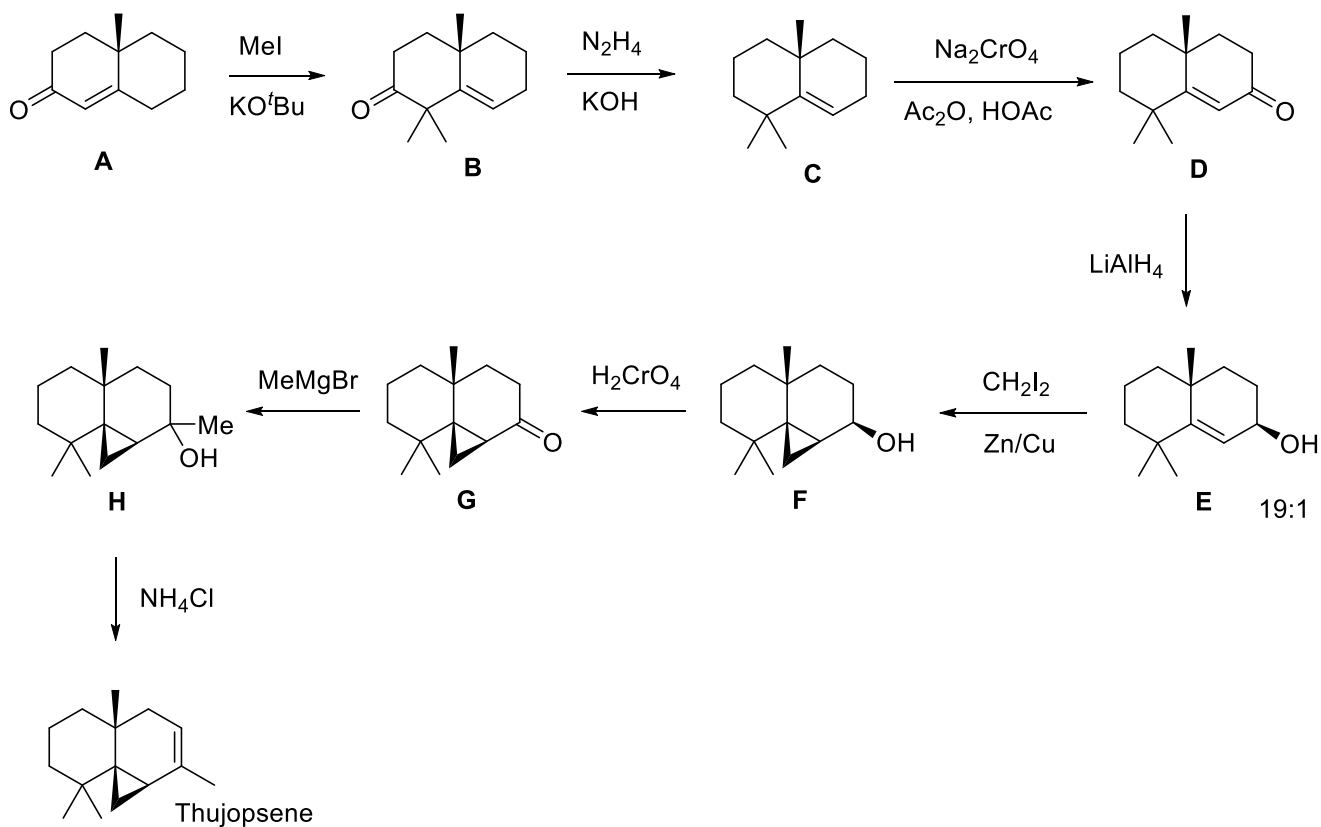


In another experiment, a volume of 200 cm³ from the initial solution was electrolyzed. Due to secondary reactions, the efficiency of the electrolysis is 90% for the metal in consideration. The three metals were deposited quantitatively in 14,50 min after applying to the solution a current of 2 A.

22.1. Balance the chemical equations and calculate the composition of the alloy in mass %.

Problem 23. Thujopsene

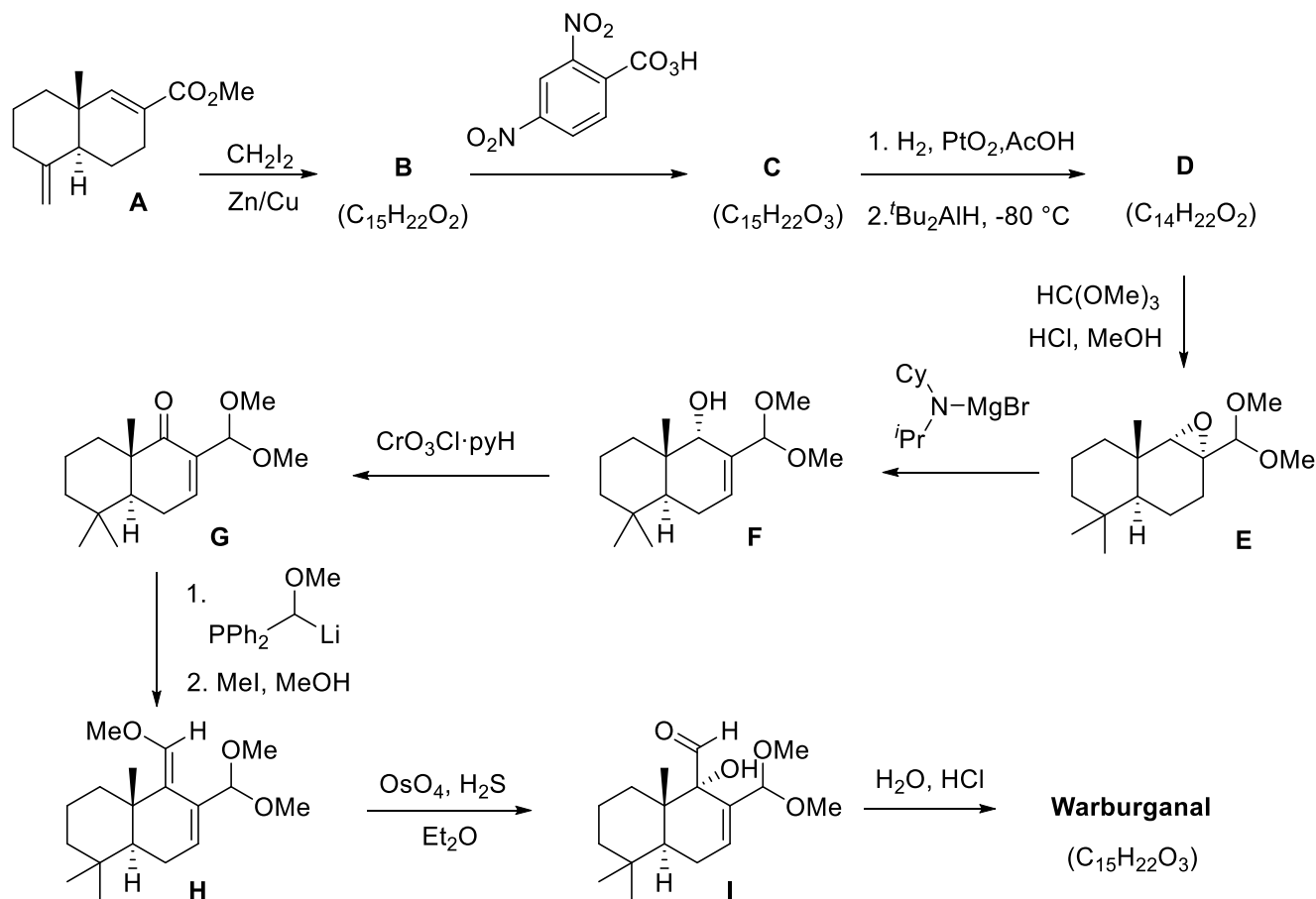
Thujopsene is a natural chemical compound, classified as a sesquiterpene, with the molecular formula C₁₅H₂₄. Thujopsene is found in essential oils of a variety of conifers, in particular, *Juniperus cedrus* and *Thujopsis dolabrata* in which it is about 2,2% mass of duramen, the woody part of the trunk of the plant. This compound may be obtained according to the following synthetic sequence (Dauben, W. G.; Ashcraft, A. C., *J. A. Chem. Soc.* **1963**, 85, 3673).



- 23.1. How many stereoisomers are possible for thujopsene?
- 23.2. Write the absolute configuration of all asymmetric carbons of **F**.
- 23.3. Why is alcohol **E** the major stereoisomer from the reduction of **D**?
- 23.4. Why is the formation of **F** stereoselective?
- 23.5. Propose a plausible mechanism for each of the chemical reactions from the sequence.
- 23.6. Propose a way to obtain the starting material **A**, from cyclohexanone and methylvinylketone.

Problem 24. Warburganal

Warburganal is a sesquiterpene of diamane kind with a potent cytotoxic activity, antifeedant, antibiotic and molluscicide isolated from leaves of the water pepper *Polygonum hydropiper*. In the laboratory, it can be obtained through a sequence involving two consecutive cycloaddition reactions (Wender, P. A.; Eck, S. L., *Tetrahedron Lett.* **1982**, 23, 1871).



24.1. Find the structures of **B**, **C**, **D** and warburganal.

24.2. Propose mechanisms for the formation of **E**, **F**, **G**, **H**, **I** and warburganal.

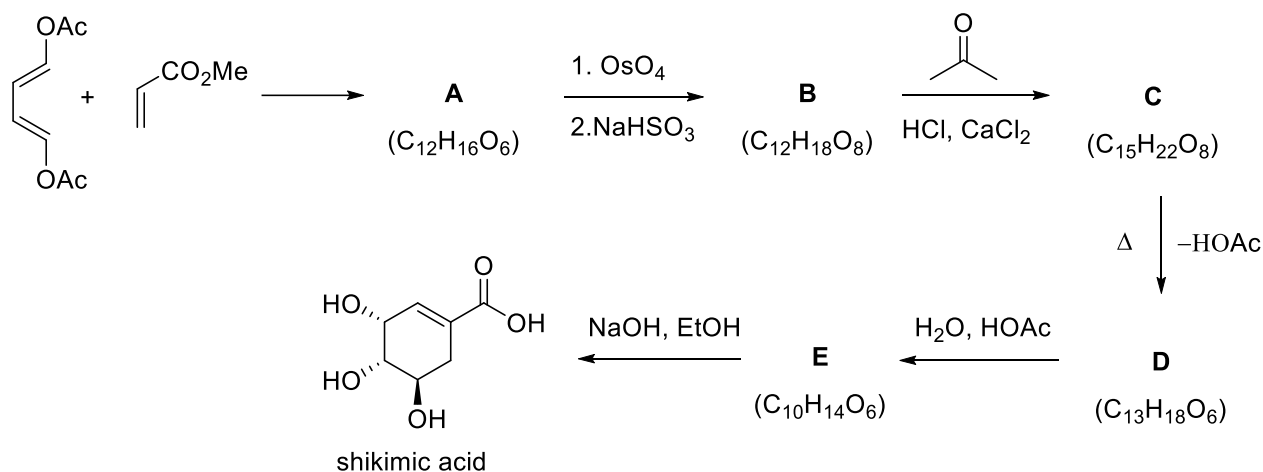
24.3. Why is the hydrogenation of **C** possible?

24.4. During the formation of **B** and **I**, only one of the two alkenes in the structure reacts. Considering these results, what generalization can you formulate about the reaction of alkenes with electrophiles?

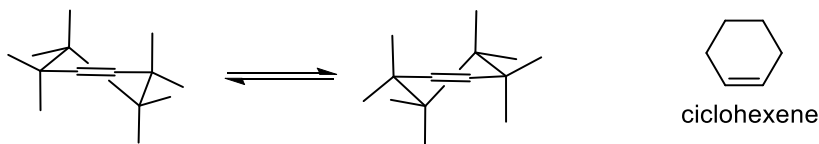
24.5. Name the reagent employed to transform **B** in **C** according to IUPAC rules.

Problem 25. Shikimic acid

Shikimic acid is a biochemical intermediate of relevance in plants and microorganisms. Its name derives from the Japanese shichimi flower (*Illicium anisatum*), the natural source from which it was extracted for the first time. In the industry, it is obtained from Chinese star anise or common anise (*Illicium verum*) and it is used for the production of the antiviral oseltamivir. The total synthesis of shikimic acid (Smisman, E. E.; Suh, J. T.; Oxman, N.; Daniels, R., *J. Am. Chem. Soc.* **1962**, *84*, 1040-1041) was reported in the middle of the last century and it is shown below.



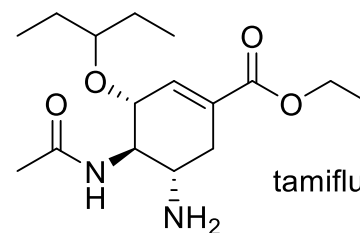
- 25.1. Name the reaction for the preparation of **A**. Characterize this reaction according to the stereochemistry, regio- and stereoselectivity.
- 25.2. In fact, during the synthesis two stereoisomers of shikimic acid are obtained. Why?
- 25.3. Name shikimic acid according to the nomenclature rules of IUPAC.
- 25.4. It is known that cyclohexene and its derivatives mostly adopt a semi-chair conformation, in which substituents are occupying pseudo-axial and pseudo-equatorial positions. Represent the least energetic conformation of **A**.



- 25.5. Find the structures of **A**, **B**, **C**, **D** and **E**.
- 25.6. Which are the products of the reaction of shikimic acid with periodic acid?

Problem 26. Tamiflu²³

Tamiflu is the anti-influenza drug currently being used to treat so-called ‘Bird-flu’. Its structure is shown on the right. The bold wedged bond is coming out of the plane of the paper, the hashed bonds are going into the plane.

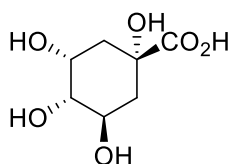


In order to increase its solubility, Tamiflu is actually sold as the phosphate salt.

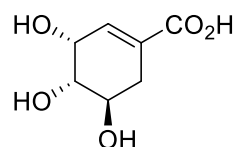
26.1. On the structure in your answer booklet, circle the atom which will be protonated in the salt.

26.2. One of the problems in the synthesis of the drug is ensuring the correct stereochemistry at each of the stereogenic (chiral) centers in the structure. On the structure in your answer booklet, mark with an asterisk each chiral center. What is the total number of stereoisomers possible for this structure?

Two syntheses of Tamiflu which have been tried start from the naturally occurring products Quinic acid and Shikimic acid.



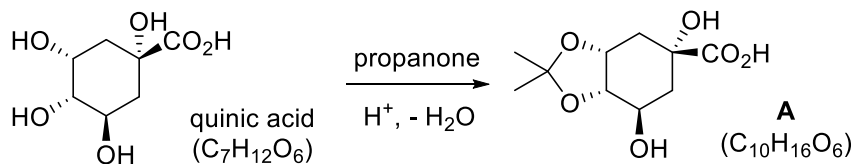
quinic acid
(C₇H₁₂O₆)



shikimic acid
(C₇H₁₀O₅)

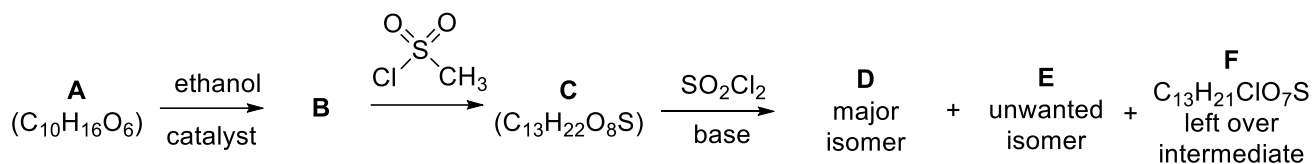
Shikimic acid has the advantage of already having the double bond in place but quinic acid has the advantage of being much cheaper.

One of the problems using either acid is trying to control which of the –OH groups reacts in a given step. The first step starting from quinic acid is therefore to protect the two –OH groups on the same side of the ring on adjacent carbon atoms by forming a so-called acetal group by the acid-catalyzed reaction with propanone:

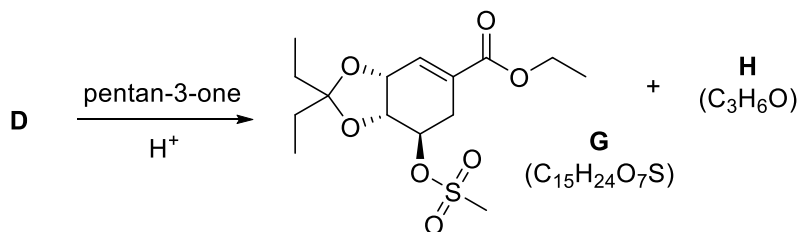


The next steps in the synthesis are outlined below. Inorganic by-products are not shown in the scheme.

²³ England Chemistry Olympiad, 2007. Theoretical Exam. Problem 5. (Adaptation)

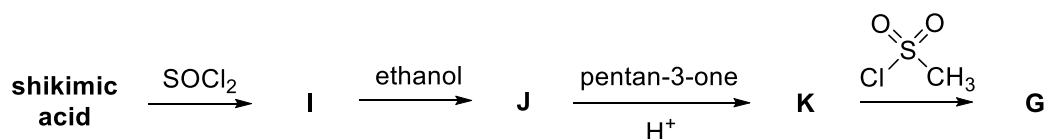


D reacts with an excess of pentan-3-one to give **G** as shown below.



26.3. Draw the structures of compounds **B**, **C**, **D**, **E** and **F** and for the by-product **H**, showing the correct stereochemistry by the use of bold and hashed bonds as appropriate.

G may easily be prepared from the more expensive Shikimic acid as outlined below. Inorganic by-products are not shown.

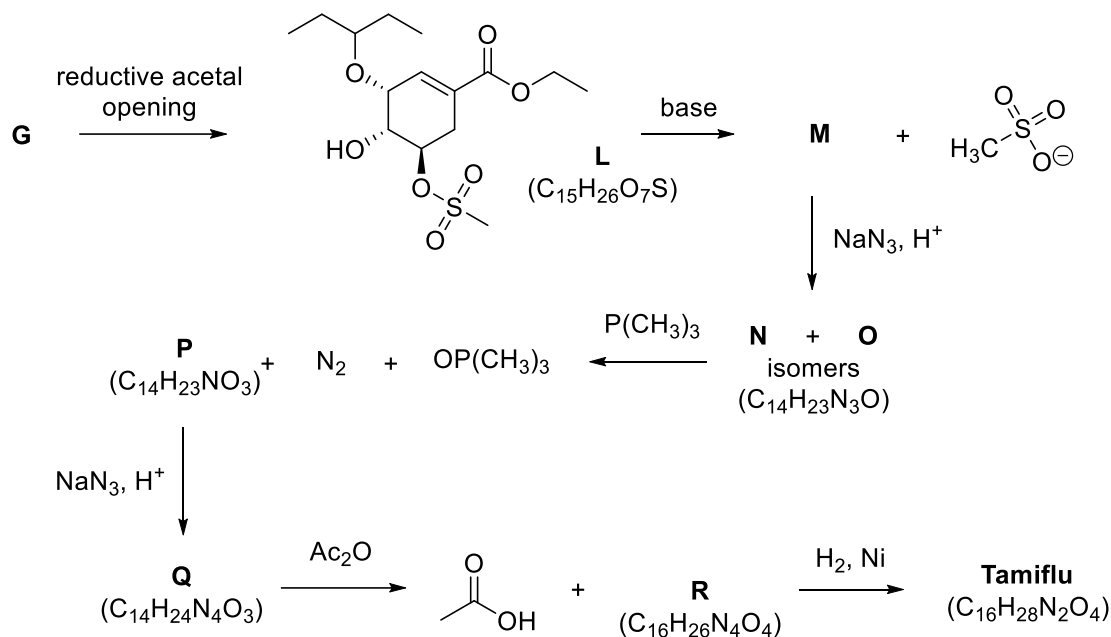


26.4. Draw the structures of compounds **I**, **J**, and **K** showing the correct stereochemistry as appropriate.

The acetal in **G** may be opened up under reducing conditions to give mainly **L** (shown below). On treatment with base **L** cyclizes to give **M** which contains a 3-membered ring. The new ring is opened up with sodium azide yielding a mix of isomers, **N** and **O** either of which yields the same cyclic compound **P** on treatment with trimethylphosphine. Further treatment with sodium azide opens the new ring to give mainly **Q** which upon acylation and hydrogenation finally yields neutral Tamiflu as outlined below.

26.5. On the structure in your answer booklet, circle the most acidic proton in **L** which is removed by the base.

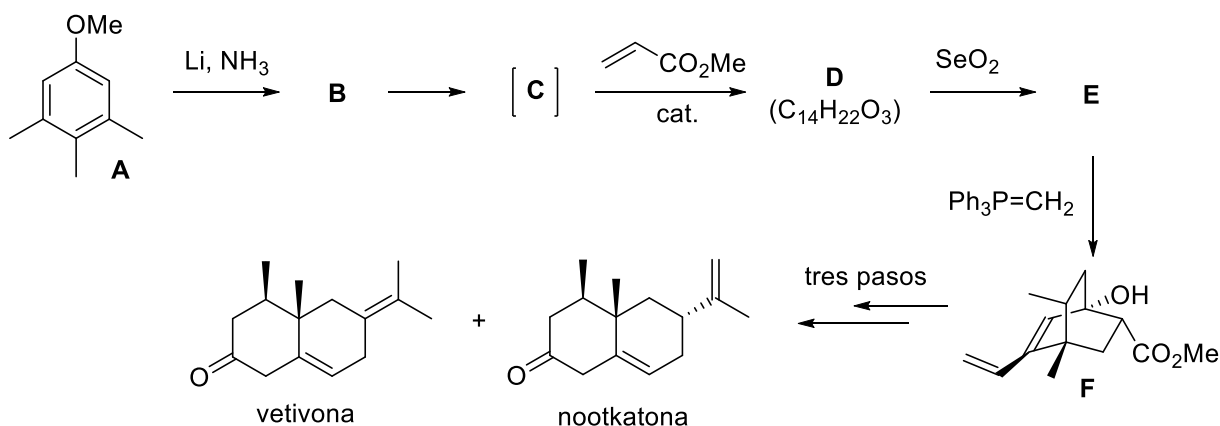
26.6. Draw the structures of compounds **M**, **N**, **O**, **P**, **Q** and **R** again showing the correct stereochemistry as appropriate.



Problem 27. Organic synthesis

Fragments of synthetic sequences employed to obtain natural products are depicted below. Your task is to find the missing compounds.

Nootkatone synthesis (Dastur, K. P. J., *J. Am. Chem. Soc.* **1974**, 95, 2605-2608)

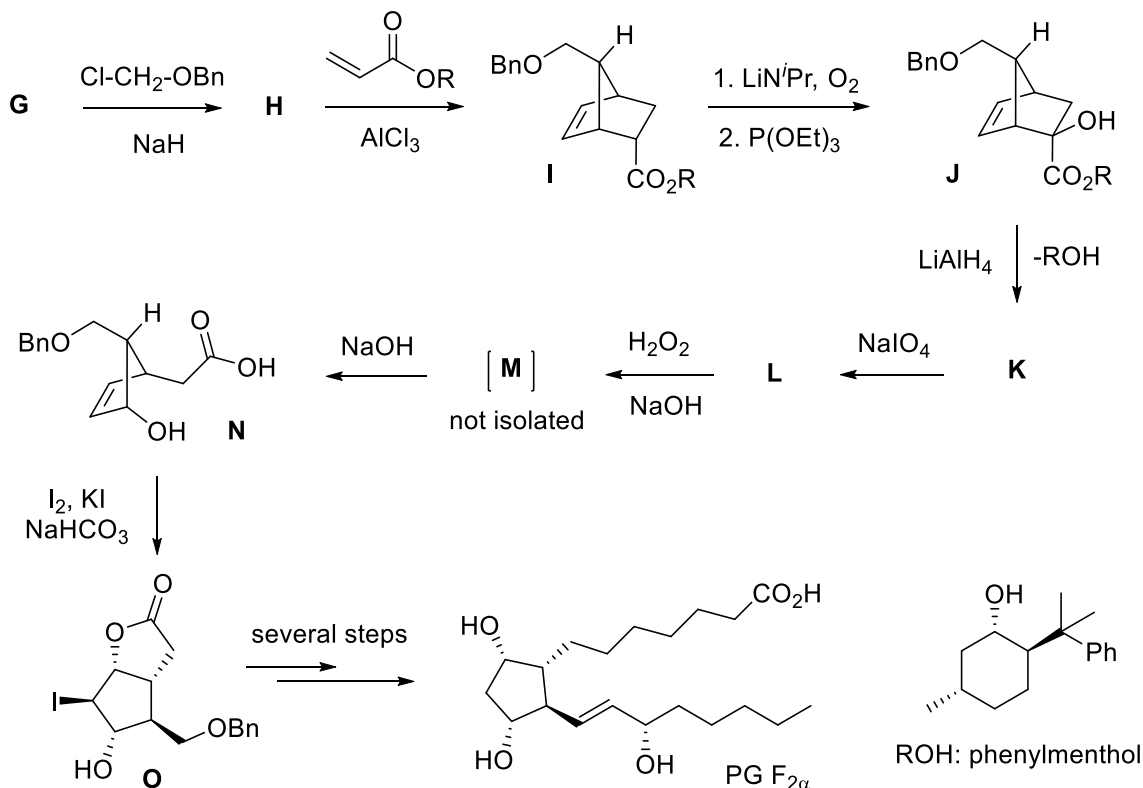


27.1. About the first reaction state: name, what other metal can be employed and what other systems can react under similar conditions.

27.2. Find the structures of **B**, **C**, **D** and **E**, if is known that before **B** is able to react with methyl acrylate, the most stable isomer **C** must form first.

27.3. Propose a mechanism for the oxidation reaction of **D** into **E**.

Prostaglandin synthesis (Corey, E. J; Ensley, H. E., *J. Am. Chem. Soc.* **1975**, 97, 6908)



27.4. Draw the structures of **G**, **H**, **K**, **L** and **M**.

27.5. Why can compound **G** react with a strong base?

27.6. At room temperature **G** exists in the form of dimer **G'** as result of an intermolecular cycloaddition reaction. Propose a structure for **G'**.

27.7. Why is it necessary to use phenylmenthol in the synthesis?

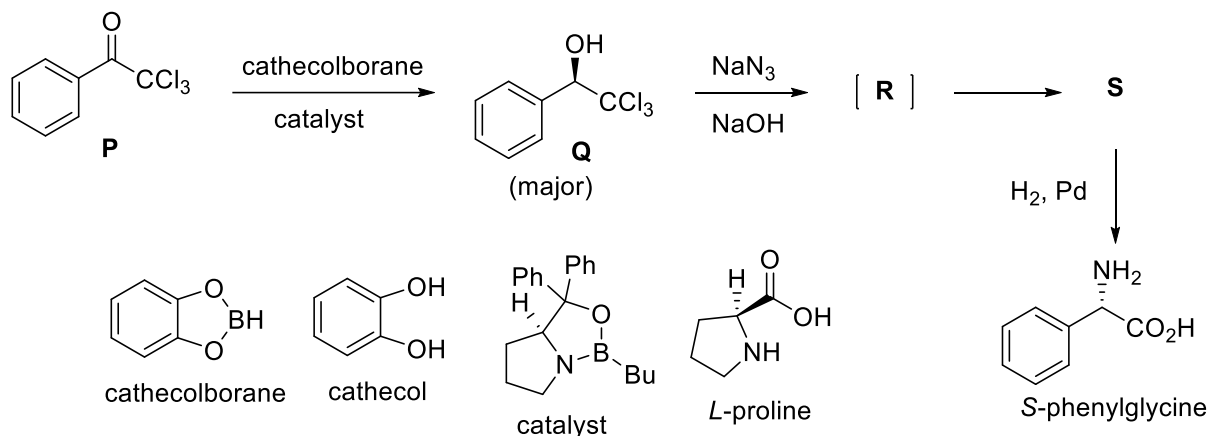
27.8. Represent the most stable chair conformation of phenylmenthol.

27.9. What is the name of the reaction to formation **M** and why this compound cannot be isolated under the conditions employed?

27.10. Propose the mechanism for the formation of **J** and **O**.

27.11. How many stereoisomers are possible for compound **O**?

27.12. Assign the absolute configuration of the asymmetric carbons of the final product.

Synthesis of α -aminoacids (Corey, E. J; Link, J. O., *J. Am. Chem. Soc.* **1992**, 114, 1906-1908)

27.13. When the starting ketone **P** is treated with hot sodium hydroxide a degradation reaction occurs. Name the reaction and write the corresponding mechanism.

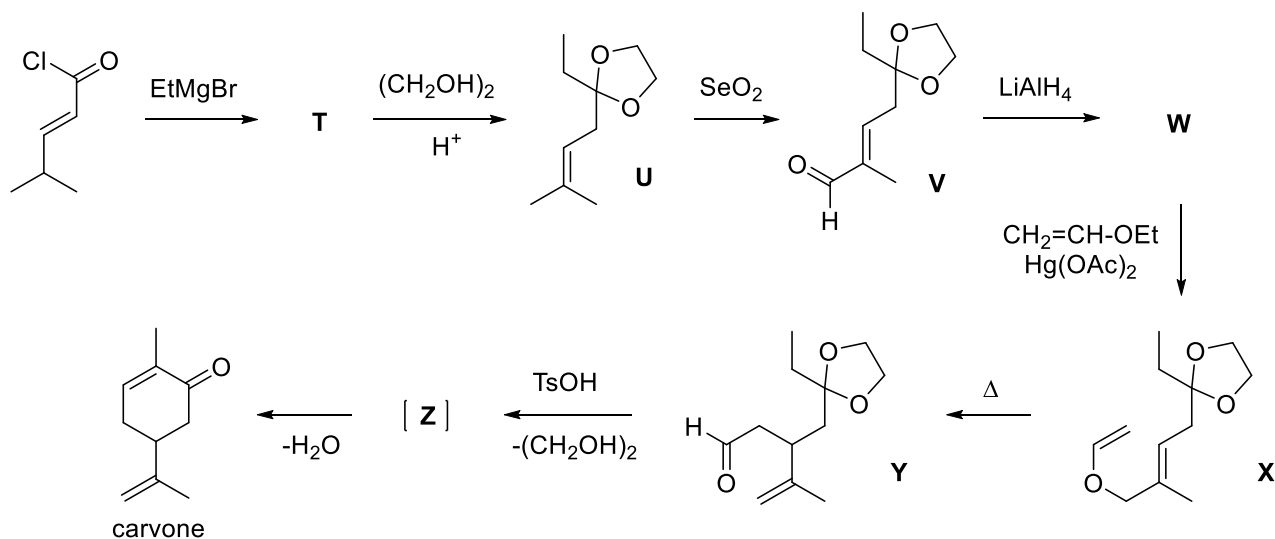
27.14. How could you classify the reduction reaction of **P** into **Q** according to its stereochemistry?

27.15. Propose structures for **R** and **S**, if it is known that **R** presents two rings in its structure.

27.16. Suggest a mechanism for the formation of **S** from **Q**.

27.17. How it is possible to obtain catecholborane and the catalyst employed starting from catechol and L-proline amino acid, respectively

27.18. What is the function of the catalyst used?

Carvone synthesis (Vid, O. P.; Sharma, S. D.; Chander, S.; Raj, I., *Ind. J. Chem.* **1966**, 4, 275)

27.19. Find the structures of **T**, **W** and **Z**.

- 27.20.** What is the name of the last reaction in the sequence?
- 27.21.** What is employed the Hg(II) salt during the formation of the vinyl ether **X**?
- 27.22.** What kind of reaction is the transformation of **X** into **Y**?

ANSWERS

Problem 1

1.1.

$$n(\text{O}_2) = \frac{PV}{RT} = \frac{390,4 \cdot 2}{8,314 \cdot 298,15} = 0,315 \text{ mol}$$

$$m(\text{O}_2) = 0,315 \cdot 32 = 10,08 \text{ g}$$

$$m(\text{O}_2) + m(\text{A}) = 10,08 + 4,71 = 14,79 \text{ g}$$

The products of incomplete combustion are C, CO, CO₂ and H₂O.

$$14,79 = m(\text{C}) + m(\text{CO}) + m(\text{CO}_2) + m(\text{H}_2\text{O}) \quad \text{because of the Law of conservation of mass}$$

$$m(\text{CO}) = 14,79 - m(\text{C}) - m(\text{CO}_2) - m(\text{H}_2\text{O})$$

$$m(\text{CO}) = 14,79 - 0,240 - 11,0 - 2,70 = 0,85 \text{ g}$$

$$n(\text{CO}) = \frac{0,85}{12 + 16} = 0,0304 \text{ mol}$$

$$n(\text{C}) = \frac{0,240}{12} = 0,0200 \text{ mol}$$

$$n(\text{CO}_2) = \frac{11,0}{12 + 2 \cdot 16} = 0,0250 \text{ mol}$$

$$n(\text{H}_2\text{O}) = \frac{2,70}{2 \cdot 1 + 16} = 0,0150 \text{ mol}$$

$$n(\text{C})_{\text{A}} = n(\text{C}) + n(\text{CO}) + n(\text{CO}_2) = 0,020 + 0,0304 + 0,25 = 0,300 \text{ mol}$$

$$n(\text{H})_{\text{A}} = 2 \cdot n(\text{H}_2\text{O}) = 2 \cdot 0,0150 = 0,300 \text{ mol}$$

$$n(\text{O})_{\text{A}} = n(\text{H}_2\text{O}) + n(\text{CO}) + 2 \cdot n(\text{CO}_2) - 2 \cdot n(\text{O}_2)$$

$$n(\text{O})_{\text{A}} = 0,150 + 0,0304 + 2 \cdot 0,250 - 2 \cdot 0,315 = 0,0504 \text{ mol}$$

$$n(\text{C}):n(\text{H}):n(\text{O}) = 0,300:0,300:0,0504 = 6:6:1 \rightarrow \text{C}_6\text{H}_6\text{O}$$

1.2.

As B is the superior homolog of C,

Let be $C = C_aH_{2a+2}$ y $B = C_{a+1}H_{2a+4}$

Combustion of C: $C_aH_{2a+2}(l) \rightarrow aCO_2(g) + (a + 1)H_2O(l)$

Combustion of B: $C_{a+1}H_{2a+4}(l) \rightarrow (a + 1)CO_2(g) + (a + 2)H_2O(l)$

$$n(CO_2) = \frac{61,6}{12 + 2 \cdot 16} = 1,40 \text{ mol}$$

$$n(H_2O) = \frac{29,7}{2 \cdot 1 + 16} = 1,65 \text{ mol}$$

$$\left. \begin{aligned} n(H_2O) &= (a + 2) \cdot n(B) + (a + 1) \cdot n(C) \\ n(CO_2) &= (a + 1) \cdot n(B) + a \cdot n(C) \end{aligned} \right\}$$

Answer 1

$$\frac{n(H_2O)}{n(CO_2)} = \frac{(a + 2) \cdot n(B) + (a + 1) \cdot n(C)}{(a + 1) \cdot n(B) + a \cdot n(C)}$$

Sustituting $n(B) = 3/2 \cdot n(C)$

$$\frac{1,65}{1,40} = \frac{(a + 2) \cdot 1,5 \cdot n(C) + (a + 1) \cdot n(C)}{(a + 1) \cdot 1,5 \cdot n(C) + a \cdot n(C)}$$

$$1,18 = \frac{1,5a + 3 + a + 1}{1,5a + 1,5 + a} = \frac{2,5a + 4}{2,5a + 1,5}$$

$$2,95a + 1,77 = 2,5a + 4 \rightarrow a = 4,96 \approx 5$$



Answer 2

Sustituting $n(B) = 3/2 \cdot n(C)$

$$\left. \begin{aligned} 1,65 &= (a + 2) \cdot 3/2 \cdot n(C) + (a + 1) \cdot n(C) \\ 1,40 &= (a + 1) \cdot 3/2 \cdot n(C) + a \cdot n(C) \end{aligned} \right\}$$

$$\left. \begin{aligned} 1,65 &= 2,5 \cdot a \cdot n(C) + 4 \cdot n(C) \\ 1,40 &= 2,5 \cdot a \cdot n(C) + 1,5 \cdot n(C) \end{aligned} \right\}$$

Subtracting the second equation from the first one, we get

$$0,25 = 2,5 \cdot n(\text{C}) \rightarrow n(\text{C}) = 0,1 \text{ mol}$$

Substituting in the first equation,

$$1,65 = 2,5 \cdot a \cdot 0,1 + 4 \cdot 0,1$$

$$1,65 = 0,25 \cdot a + 0,4 \rightarrow a = 5$$



Problem 2

2.1.

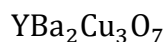
In 100 g of superconductor **A**,

$$n(\text{Y}) = \frac{m(\text{Y})}{A_r(\text{Y})} = \frac{13,4}{88,91} = 0,151 \text{ mol}$$

$$n(\text{Ba}) = \frac{m(\text{Ba})}{A_r(\text{Ba})} = \frac{41,2}{137,3} = 0,300 \text{ mol}$$

$$n(\text{Cu}) = \frac{m(\text{Cu})}{A_r(\text{Cu})} = \frac{28,6}{63,5} = 0,450 \text{ mol}$$

$$n(\text{O}) = \frac{m(\text{O})}{A_r(\text{O})} = \frac{100 - 13,4 - 41,2 - 28,6}{16,0} = 1,05 \text{ mol}$$



2.2.

The average oxidation number of copper is $+7/3$, meaning that per every two Cu(II) ions in the mixt oxide, there is one Cu(III) ion.

2.3.

The formula of **B** is $\text{YBa}_2\text{Cu}_3\text{O}_{6.5}$.

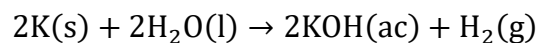
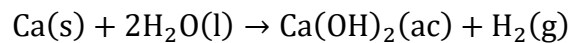
$$M(\text{A}) = 666,19 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{B}) = 658,19 \text{ g} \cdot \text{mol}^{-1}$$

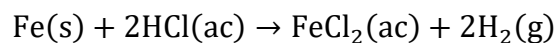
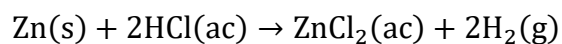
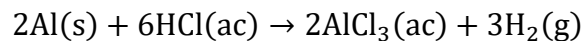
$$m(\text{B}) = M(\text{B}) \cdot \frac{m(\text{A})}{M(\text{A})} = 658,19 \cdot \frac{84,2}{666,19} = 83,19 \text{ mg}$$

Problem 3

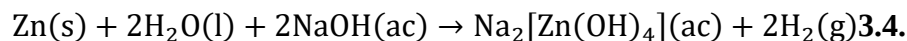
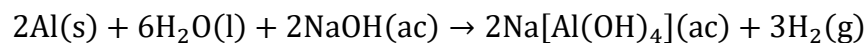
3.1.



3.2.



3.3.

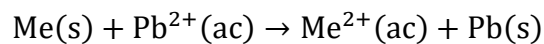
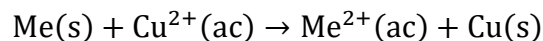


3.4.



3.5.

$$n(\text{Me}) = n(\text{Cu}) = n(\text{Pb})$$



$$100 \text{ g} - n(\text{Me}) \cdot M(\text{Me}) + n(\text{Me}) \cdot 207,2 = 119 \text{ g}$$

$$100 \text{ g} - n(\text{Me}) \cdot M(\text{Me}) + n(\text{Me}) \cdot 63,55 = 90,2 \text{ g}$$

$$9,8 \text{ g} + n(\text{Me}) \cdot 63,55 = -19 \text{ g} + n(\text{Me}) \cdot 207,2$$

$$n(\text{Me}) = 0,2 \text{ mol} \quad \rightarrow \quad M(\text{Me}) = 112,4 \text{ g} \cdot \text{mol}^{-1} \quad \rightarrow \quad \text{Me} = \text{Cd}$$

Problem 4

4.1.

The formula of the hydrogen compound can be written as $\mathbf{AH}_x$.

$$\omega(\text{H}) = \frac{x \cdot M(\text{H})}{M(\mathbf{A}) + x \cdot M(\text{H})} \rightarrow M(\mathbf{A}) = x \cdot M(\text{H}) \cdot \left(\frac{1}{\omega(\text{H})} - 1 \right)$$

In compound **D**

$$M(\mathbf{A}) = x \cdot 1,01 \cdot \left(\frac{1}{0,112} - 1 \right) = x \cdot 8,01$$

Si $x = 2 \rightarrow M(\mathbf{A}) = 16,02$ oxygen and OH_2

Si $x = 4 \rightarrow M(\mathbf{A}) = 32,03$ sulfur and SH_4 ?

Other options are ignored because **A** is a non-metal.

In compound **E**

$$M(\mathbf{A}) = x \cdot 1,01 \cdot \left(\frac{1}{0,059} - 1 \right) = x \cdot 16,1$$

Si $x = 1 \rightarrow M(\mathbf{A}) = 16,1$ oxygen and OH

Si $x = 2 \rightarrow M(\mathbf{A}) = 32,2$ sulfur and SH_2

Si $x = 5 \rightarrow M(\mathbf{A}) = 80,5$ bromine and BrH_5 ?

Because **A** exists in the form of gaseous simple substances, non-metal **A** is oxygen.

A = O **B** = O_2 **C** = O_3 **D** = H_2O **E** = H_2O_2

4.2.

The formula of the hydrogen compound can be written as $\mathbf{FH}_x$.

$$\omega(\text{H}) = \frac{x \cdot M(\text{H})}{M(\mathbf{F}) + x \cdot M(\text{H})} \rightarrow M(\mathbf{F}) = x \cdot M(\text{H}) \cdot \left(\frac{1}{\omega(\text{H})} - 1 \right)$$

In compound **H**

$$M(\mathbf{F}) = x \cdot 1,01 \cdot \left(\frac{1}{0,178} - 1 \right) = x \cdot 8,01$$

Si $x = 3 \rightarrow M(\mathbf{F}) = 13,99$ nitrogen and NH_3

Si $x = 7 \rightarrow M(F) = 32,65$ sulfure and SH_7 ?

In compound **I**

$$M(F) = x \cdot 1,01 \cdot \left(\frac{1}{0,126} - 1 \right) = x \cdot 16,1$$

Si $x = 2 \rightarrow M(F) = 14,01$ nitrogen and NH_2

Si $x = 4 \rightarrow M(F) = 28,02$ silicon and SiH_4

Non-metal **F** is nitrogen.

F = N **G** = N_2 **H** = NH_3 **I** = N_2H_4

4.3.

The formula of these compounds can be written as JO_x , where x may be a fraction.

$$\omega(J) = \frac{M(J)}{M(J) + x \cdot M(O)} \rightarrow M(J) = x \cdot M(O) \cdot \left(\frac{\omega(J)}{1 - \omega(J)} \right) \quad y \quad x = \frac{M(J)}{M(O)} \cdot \frac{1 - \omega(J)}{\omega(J)}$$

In compound **K**

$$M(J) = x \cdot 16,00 \cdot \left(\frac{0,590}{1 - 0,590} \right) = x \cdot 23,02$$

Si $x = 1 \rightarrow M(J) = 23,02$ sodium and NaO

Si $x = 3,5 \rightarrow M(J) = 80,59$ bromine? and Br_2O_7 ?

In compound **L**

$$M(J) = x \cdot 16,00 \cdot \left(\frac{0,742}{1 - 0,742} \right) = x \cdot 46,02$$

Si $x = 0,5 \rightarrow M(J) = 23,01$ sodium and Na_2O

Si $x = 1,5 \rightarrow M(J) = 69,02$ gallium? and Ga_2O_3

Metal **J** sodium can form oxide and peroxide.

In compound **M**

$$x = \frac{22,99}{16,00} \cdot \frac{1 - 0,324}{0,324} = 3$$

It is sodium ozonide, NaO_3 .

In compound **L**

$$x = \frac{22,99}{16,00} \cdot \frac{1 - 0,418}{0,418} = 2$$

It is sodium superoxide, NaO_2 .

J = Na **K** = Na_2O_2 **L** = Na_2O **M** = NaO_3 **N** = NaO_2

4.4.

The formula of these compounds can be written as JN_x , where x may be a fraction.

$$\omega(\text{J}) = \frac{M(\text{J})}{M(\text{J}) + x \cdot M(\text{N})} \rightarrow x = \frac{M(\text{J})}{M(\text{N})} \cdot \frac{1 - \omega(\text{J})}{\omega(\text{J})}$$

In compound **O**

$$x = \frac{22,99}{14,01} \cdot \frac{1 - 0,354}{0,354} = 3$$

It is sodium azide, NaN_3 .

In compound **Q**

$$x = \frac{22,99}{14,01} \cdot \frac{1 - 0,831}{0,831} = 0,334 = \frac{1}{3}$$

It is sodium nitride, Na_3N .

In compound **P**

$$x = \frac{22,99}{16,00} \cdot \frac{1 - 0,589}{0,589} = 1,15$$

This result suggests the presence of another chemical element in the structure. Because **P** is obtained from NH_3 , it should contain H.

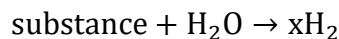
Testing for **P** = NaNH_2

$$\omega(\text{Na}) = \frac{22,99}{22,99 + 2 \cdot 1,01 + 14,01} = 0,589$$

O = NaN_3 **P** = NaNH_2 **Q** = Na_3N

4.5.

The released gas is hydrogen.



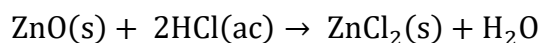
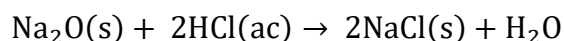
$$n(\text{H}_2) = \frac{V}{V_m} = \frac{1,13}{22,4} = 0,0504 \text{ mol}$$

$$M(\text{substance}) = \frac{m}{n(\text{H}_2)} \cdot x = \frac{1,21}{0,504} \cdot x = x \cdot 24,01 \text{ g} \cdot \text{mol}^{-1}$$

$$\text{Para } x = 1 \rightarrow M(\text{substance}) = 24,01 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{substance}) - M(\text{Na}) = 24,01 - 22,99 = 1,02 \approx M(\text{H}) \rightarrow \text{NaH}$$

4.6.



$$n(\text{HCl}) = c(\text{HCl}) \cdot V(\text{HCl}) = 1,00 \cdot 0,150 = 0,150 \text{ mol}$$

$$M(\text{Na}_2\text{O}) = 2 \cdot 22,99 + 16,00 = 61,98 \text{ g} \cdot \text{mol}^{-1} \quad M(\text{ZnO}) = 65,41 + 16,00 = 81,41 \text{ g} \cdot \text{mol}^{-1}$$

$$m(\text{Na}_2\text{O}) + m(\text{ZnO}) = 7,17$$

$$\left. \begin{array}{l} 61,98 \cdot n(\text{Na}_2\text{O}) + 81,41 \cdot n(\text{ZnO}) = 7,17 \\ 2 \cdot n(\text{Na}_2\text{O}) + 2 \cdot n(\text{ZnO}) = 0,200 / -30,99 \end{array} \right\}$$

$$19,43 \cdot n(\text{ZnO}) = 0,972 \rightarrow n(\text{ZnO}) = 0,972/19,43 = 0,0500 \text{ mol}$$

$$m(\text{ZnO}) = 81,41 \cdot 0,500 = 4,07 \text{ g}$$

$$m(\text{Na}_2\text{O}) = 7,17 - 4,07 = 3,10 \text{ g}$$

$$\omega_{\%}(\text{Na}_2\text{O}) = 4,07/7,17 \cdot 100 = 56,8 \%$$

$$\omega_{\%}(\text{ZnO}) = 100 - 56,8 \% = 43,2\%$$

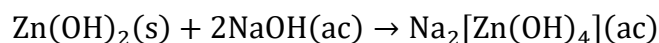
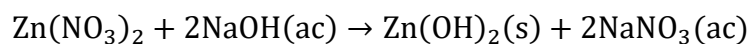
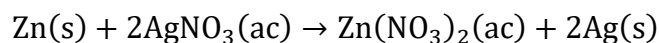
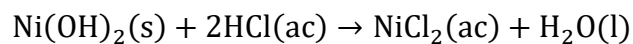
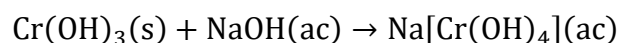
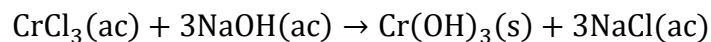
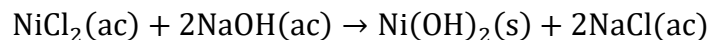
Problem 5

5.1.

Mixture A: NiCl_2 y CrCl_3

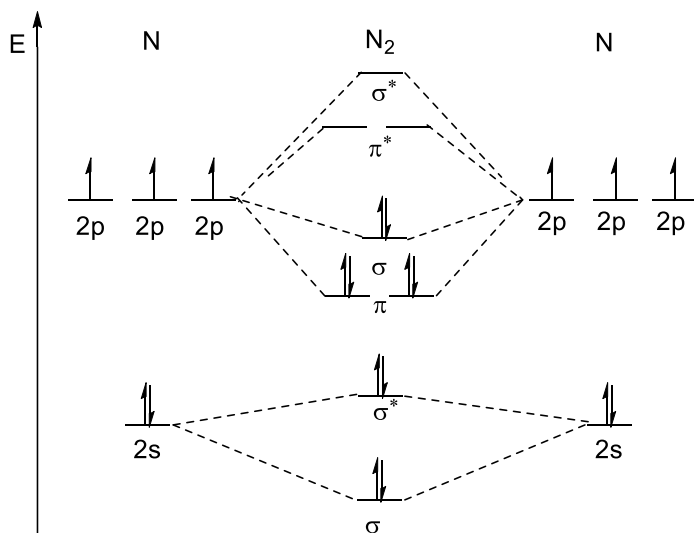
Mixture B: Zn y AgNO_3

5.2.



Problem 6

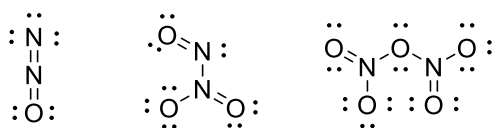
6.1.



6.2.

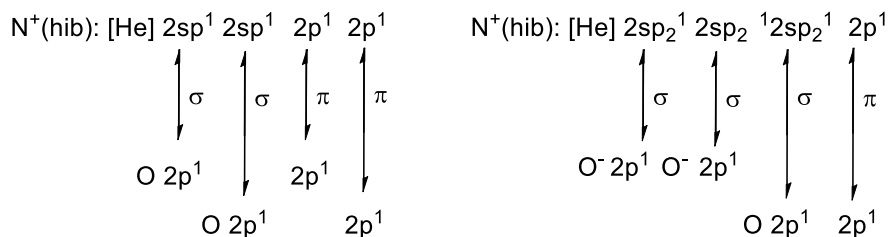
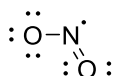
The bond order for N_2 molecule is 3, while for N_2^+ ion the bond order is 2,5; therefore, the bonding in the neutral molecule is stronger and its dissociation is harder.

6.3.

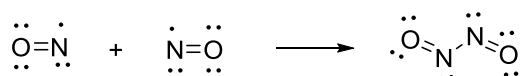


6.4.

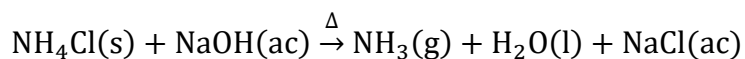
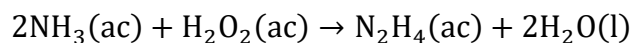
The ions present in solid N_2O_5 are NO_2^+ and NO_3^- .

**6.5.**

The central nitrogen atom has three electronic domains, and one domain is occupied by a single electron, non-bonded to any atom. Hence, the geometry for this molecule is angular. The angle for the ONO bond is about 120° .

6.6.

Nitrogen monoxide is a highly reactive radical because it has an unpaired electron. At low temperatures, two NO molecules combine to form the dimer, in which all the electrons are paired. The weak N-N bond is broken at high temperatures, making the monomer more stable.

6.7.**6.8.****6.9.**

Nitrogen is in period two and has only four atomic orbitals in its last level, so the maximum number of bonds it can form is 4. On the other hand, phosphorus is in period three and has empty d orbitals that can accept electrons to form more bonds.

6.10.

Ammonium nitrate is very soluble in water due to several factors: it is an ionic substance with low lattice energy due to the large size of its ions, when it interacts with water, its ions form hydrogen bonds, and the ammonium ion dissociates to form ammonia.

Problem 7**7.1.**

Three moles of ions

7.2.

Methane

7.3.

14/3

7.4.

10 mol of calcium

7.5.

Al_2O_3

7.6.

AlCl_3 o Al_2Cl_6

7.7.

sp_3d

7.8.

$\text{C}_9\text{H}_8\text{O}_4$

7.9.

8,3,3,2,4,3

7.10.

6,78 g

7.11.

Cs

7.12.

Ne y F^-

7.13.

Chlorine atoms have one electron less than chloride.

7.14.

Four electrons

7.15.

Ten electrons

7.16.

$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$

7.17.

Ca

7.18.

Mg^{2+}

7.19.

CO_3

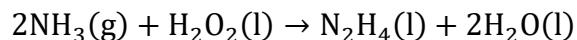
7.20.

Trigonal pyramidal

Problem 8

8.1.

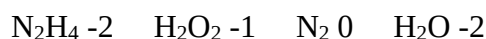
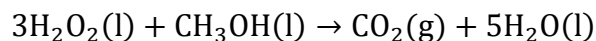


8.2.

$$\Delta H_{\text{r}}^\circ = \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4) + 2 \cdot \Delta H_{\text{f}}^\circ(\text{H}_2\text{O}) - 2 \cdot \Delta H_{\text{f}}^\circ(\text{NH}_3) - \Delta H_{\text{f}}^\circ(\text{H}_2\text{O}_2)$$

$$241 = \Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4) + 2 \cdot (-285,8) - 2 \cdot (-46,1) - (-187,8)$$

$$\Delta H_{\text{f}}^\circ(\text{N}_2\text{H}_4) = 532,6 \text{ kJ} \cdot \text{mol}^{-1}$$

**8.3.****8.4.****8.5.**

$$M(\text{N}_2\text{H}_4) = 2 \cdot 14 + 4 \cdot 1 = 32 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{CH}_3\text{OH}) = 12 + 4 \cdot 1 + 16 = 32 \text{ g} \cdot \text{mol}^{-1}$$

$$Q(\text{N}_2\text{H}_4) = \Delta H_{\text{comb}}^\circ \cdot \frac{\rho \cdot V}{M} = -622,2 \cdot \frac{1,021 \cdot 225000}{32} = -4,47 \cdot 10^6 \text{ kJ}$$

$$Q(\text{CH}_3\text{OH}) = \Delta H_{\text{comb}}^\circ \cdot \frac{\rho \cdot V}{M} = -726,0 \cdot \frac{0,7918 \cdot 862000}{32} = -1,55 \cdot 10^7 \text{ kJ}$$

$$Q(\text{total}) = -1,55 \cdot 10^7 + (-4,47 \cdot 10^6) = -2,00 \cdot 10^7 \text{ kJ}$$

8.6.

Gravimetric energy density

$$\text{N}_2\text{H}_4 \quad \frac{-622,2 \text{ kJ} \cdot \text{mol}^{-1}}{32 \text{ g} \cdot \text{mol}^{-1}} = -19,4 \text{ kJ} \cdot \text{g}^{-1}$$

$$\text{CH}_3\text{OH} \quad \frac{-726,0 \text{ kJ} \cdot \text{mol}^{-1}}{32 \text{ g} \cdot \text{mol}^{-1}} = -22,7 \text{ kJ} \cdot \text{g}^{-1}$$

Volumetric energy density

$$\text{N}_2\text{H}_4 \quad -19,4 \text{ kJ} \cdot \text{g}^{-1} \cdot 1,021 \text{ g} \cdot \text{mL}^{-1} = -19,8 \text{ kJ} \cdot \text{mL}^{-1}$$

$$\text{CH}_3\text{OH} \quad -22,7 \text{ kJ} \cdot \text{g}^{-1} \cdot 0,7918 \text{ g} \cdot \text{mL}^{-1} = -18,0 \text{ kJ} \cdot \text{mL}^{-1}$$

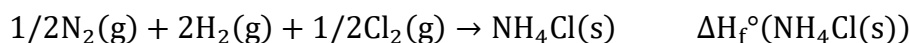
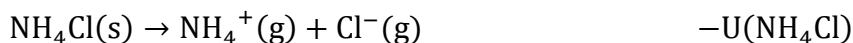
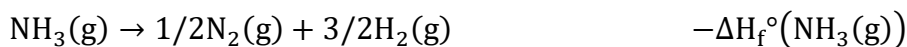
8.7.

When the limiting factor is weight, methanol is the best fuel.

When the limiting factor is volume, hydrazine is a more adequate fuel.

Problem 9

9.1.



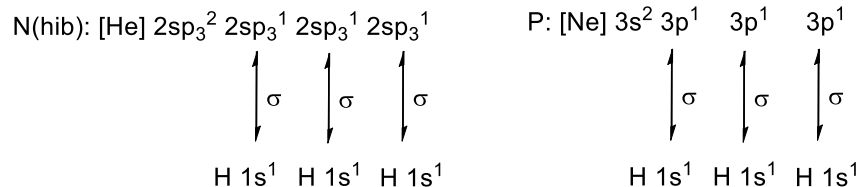
$$AP = -\Delta H_f^\circ(\text{NH}_3(\text{g})) - I(\text{H}) - U(\text{NH}_4\text{Cl}) + \Delta H_f^\circ(\text{NH}_4\text{Cl}(\text{s})) - EA(\text{Cl}) - 1/2\Delta H_{\text{dis}}^\circ(\text{Cl}_2) - 1/2\Delta H_{\text{dis}}^\circ(\text{H}_2) = 46 - 1313 + 651 - 314 + 349 - 242/2 - 431/2 = -918 \text{ kJ} \cdot \text{mol}^{-1}$$

9.2.

Ammonia is the best base because it has a larger proton affinity.

9.3.

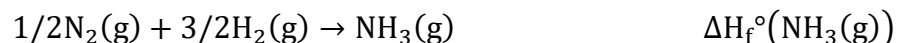


9.4.**9.5.**

N is in period two and its orbitals have similar size to the 1s orbitals of H; this allows for a stronger interaction.

9.6.

In the dinitrogen molecule, there is a triple bond, and the size of the orbitals is small; both factors cause the covalent bond to be very strong.

9.7.

$$\Delta H_f^\circ(\text{NH}_3(\text{g})) = 1/2\Delta H_{\text{dis}}^\circ(\text{N}_2) + 3/2\Delta H_{\text{dis}}^\circ(\text{H}_2) - 3\Delta H^\circ(\text{N} - \text{H})$$

$$-46 = 1/2\Delta H_{\text{dis}}^\circ(\text{N}_2) + 3/2 \cdot 431 - 3 \cdot 388$$

$$\Delta H^\circ(\text{N} - \text{N}) = 943 \text{ kJ} \cdot \text{mol}^{-1}$$

Problem 10**10.1.**

$$U(\text{FeO}) = -(416 + 759 + 1561 + 249 + 657) + (-278) = -3920 \text{ kJ} \cdot \text{mol}^{-1}$$

$$U(\text{CaO}) = -(178 + 590 + 1145 + 249 + 657) + (-635) = -3454 \text{ kJ} \cdot \text{mol}^{-1}$$

10.2.

iron(II) oxide

10.3.

$$\Delta H_r^\circ = \Delta H_f^\circ(\text{CaO}) - \Delta H_f^\circ(\text{FeO}) = -635 - (-278) = -357 \text{ kJ} \cdot \text{mol}^{-1}$$

10.4.

It is not possible because calcium is very expensive, the reaction products are challenging to separate, and the reaction is very violent.

10.5.

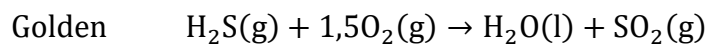
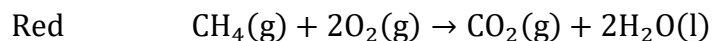
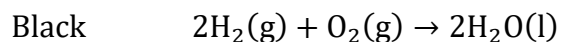
The behavior of these oxides depends on the balance between ionization energy and lattice energy.

Problem 11**11.1.**

$$n(\text{total}) = \frac{V(\text{total})}{V_m} = \frac{5,1 \cdot 10^3}{15} = 340 \text{ mol}$$

11.2.

$$n(\text{O}_2) = 0,15 \cdot 340 = 51 \text{ mol}$$

11.3.**11.4.**

In the Black Dragon

$$n(\text{H}_2) = 0,25 \cdot 340 = 85 \text{ mol} \quad n(\text{O}_2) = 0,5 \cdot n(\text{H}_2) = 0,5 \cdot 85 = 42,5 \text{ mol}$$

Additional oxygen is not needed.

In the Red Dragon

$$n(\text{CH}_4) = 0,30 \cdot 340 = 102 \text{ mol} \quad n(\text{O}_2) = 2 \cdot n(\text{H}_2) = 2 \cdot 102 = 204 \text{ mol}$$

$$n(\text{O}_2) \text{ needed} = 204 - 51 = 153 \text{ mol}$$

In the Golden Dragon

$$n(\text{H}_2\text{S}) = 0,20 \cdot 340 = 68 \text{ mol} \quad n(\text{O}_2) = 1,5 \cdot n(\text{H}_2\text{S}) = 1,5 \cdot 68 = 102 \text{ mol}$$

$$n(\text{O}_2) \text{ necesario} = 102 - 51 = 51 \text{ mol}$$

11.5.

In the Black Dragon

$$\Delta H_r^\circ = n(\text{H}_2) \cdot \Delta H_c^\circ(\text{H}_2) = 85 \cdot (-240) = -2,0 \cdot 10^4 \text{ kJ}$$

In the Red Dragon

$$\Delta H_r^\circ = n(\text{CH}_4) \cdot \Delta H_c^\circ(\text{CH}_4) = 102 \cdot (-800) = -8,2 \cdot 10^4 \text{ kJ}$$

In the Golden Dragon

$$\Delta H_r^\circ = n(\text{H}_2\text{S}) \cdot \Delta H_c^\circ(\text{H}_2\text{S}) = 68 \cdot (-520) = -3,5 \cdot 10^4 \text{ kJ}$$

Problem 12

12.1.

Reactants: Br_2, H_2 Products: HBr Intermediaries: H, Br

Molecularity: 2 Rate-determining step: Initiation

12.2.

$$\frac{d[\text{Br}]}{dt} = 2k_1[\text{Br}_2] + k_3[\text{H}][\text{Br}_2] + k_4[\text{H}][\text{HBr}] - k_2[\text{Br}][\text{H}_2] - 2k_5[\text{Br}]^2$$

$$\frac{d[\text{H}]}{dt} = k_2[\text{Br}][\text{H}_2] - k_3[\text{H}][\text{Br}_2] - k_4[\text{H}][\text{HBr}]$$

12.3.

When the reaction starts, there are only reactants,

$$\frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2]} = k[\text{H}_2][\text{Br}_2]^{1/2}$$

The partial reaction order for Br_2 is $1/2$.

When the reaction is about to end, there are practically no reactants,

$$\frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{k'[\text{HBr}]}$$

$$\frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2]} = k[\text{H}_2][\text{Br}_2]^{1/2}$$

The partial reaction order for Br_2 is $3/2$.

12.4.

$$k = 2k_2 \sqrt{\frac{k_1}{k_5}} = 2A_2 e^{-\frac{E_{a2}}{RT}} \sqrt{\frac{A_1 e^{-\frac{E_{a1}}{RT}}}{A_5 e^{-\frac{E_{a5}}{RT}}}}$$

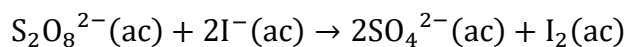
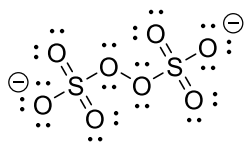
$$k = 2A_2 \sqrt{\frac{A_1}{A_5}} e^{-\frac{E_{a2}}{RT}} \sqrt{\frac{e^{-\frac{E_{a1}}{RT}}}{e^{-\frac{E_{a5}}{RT}}}} = 2A_2 \sqrt{\frac{A_1}{A_5}} e^{-\frac{E_{a2} + \frac{E_{a1}}{2} - \frac{E_{a5}}{2}}{RT}}$$

$$A_{\text{Total}} = 2A_2 \left(\frac{A_1}{A_5}\right)^{1/2}$$

$$E_{a\text{Total}} = E_{a2} + \frac{E_{a1}}{2} - \frac{E_{a5}}{2}$$

Problem 13

13.1.



13.2.

$$v = k \cdot c(\text{S}_2\text{O}_8^{2-})^a \cdot c(\text{I}^-)^b$$

In experiments 1 y 2

$$\frac{1,54 \cdot 10^{-8}}{1,10 \cdot 10^{-8}} = \frac{0,00014^a}{0,0010^a} \rightarrow a = 1$$

In experiments 1 y 3

$$\frac{2,97 \cdot 10^{-8}}{1,10 \cdot 10^{-8}} = \frac{0,015^b}{0,010^b} \rightarrow b = 1$$

Rate law: $v = k \cdot c(\text{S}_2\text{O}_8^{2-}) \cdot c(\text{I}^-)$

$$k = \frac{v}{c(\text{S}_2\text{O}_8^{2-}) \cdot c(\text{I}^-)} = \frac{1,1 \cdot 10^{-8}}{0,0001 \cdot 0,1} = 0,011 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

13.3.

$$\ln\left(\frac{k_2}{k_1}\right) = -\frac{E_A}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\left(\frac{10}{1}\right) = -\frac{42000}{8,31} \cdot \left(\frac{1}{T_2} - \frac{1}{298}\right) \rightarrow T_2 = 345 \text{ K} \approx 72 \text{ }^\circ\text{C}$$

Problem 14

14.1.

i) Correct ii) Incorrect

14.2.

-9 °C

14.3.

$$\frac{M(\text{MgCl}_2)}{M(\text{NaCl})} = \frac{24,3 + 35,5 \cdot 2}{23,0 + 35,5} = 1,63 \quad \frac{i(\text{MgCl}_2)}{i(\text{NaCl})} = \frac{3}{2} = 1,5$$

The freezing point increases.

Problem 15

15.1.

$$n(\text{total}) = \frac{p \cdot V}{R \cdot T} = \frac{71800 \cdot 100 \cdot 10^{-6}}{8,314 \cdot 298} = 2,90 \cdot 10^{-3} \text{ mol}$$

$$M(\text{C}_{21}\text{H}_{23}\text{O}_5\text{N}) = 369,45 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = 342,34 \text{ g} \cdot \text{mol}^{-1}$$

$$n_1 + n_2 = 2,90 \cdot 10^{-3} \text{ mol} \quad m_1 + m_2 = 1,00 \text{ g}$$

$$m(\text{C}_{21}\text{H}_{23}\text{O}_5\text{N}) = 0,098 \text{ g (9,8\%)}$$

$$m(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = 0,902 \text{ g (90,2\%)}$$

15.2.

Along with the sample, water is added to the solution. Crystallization removes water from the solution.

$$M(\text{CaCl}_2) = 110,98 \text{ g} \cdot \text{mol}^{-1}$$

$$M(\text{CaCl}_2 \cdot 6\text{H}_2\text{O}) = 219,1 \text{ g} \cdot \text{mol}^{-1}$$

In the residue of 74,9 g of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ there are

$$74,9 \cdot \frac{110,98}{219,1} = 37,94 \text{ g de CaCl}_2$$

$$m(\text{total}) - m(\text{residue}) = m(\text{saturated solution})$$

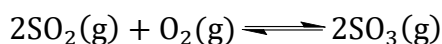
$$230,00 - 74,90 = 155,10 \text{ g} = m(\text{CaCl}_2)_{\text{dis}} + m(\text{H}_2\text{O})_{\text{dis}}$$

$$\frac{m(\text{CaCl}_2)_{\text{dis}}}{m(\text{H}_2\text{O})_{\text{dis}}} = \frac{74,5}{100} \rightarrow m(\text{CaCl}_2)_{\text{dis}} = 66,22 \text{ g} \quad m(\text{H}_2\text{O})_{\text{dis}} = 88,88 \text{ g}$$

Hence, in 150 g of the sample there is

$$66,22 + 37,94 = 104,16 \text{ g (0,939 mol) CaCl}_2$$

$$150 - 104,16 = 45,84 \text{ g (2,54 mol) H}_2\text{O} \quad \approx 2,7 \text{ mol of H}_2\text{O per 1 mol of CaCl}_2$$

Problem 16**16.1.**

$$a - 2x \quad a - x \quad 2x$$

$$n(\text{SO}_3)_{\text{eq}} = n(\text{SO}_2)_{\text{eq}} = a - 2x = 2x \rightarrow a = 4x$$

$$n(\text{total})_{\text{eq}} = (a - 2x) + (a - x) + 2x = 7x$$

$$n(\text{total})_{\text{eq}} = \frac{114,8 \cdot 4,00}{0,0821 \cdot 1000,15} = 5,60 \text{ mol} \quad x = 0,800 \text{ mol}$$

$$n(\text{SO}_2)_{\text{eq}} = n(\text{SO}_3)_{\text{eq}} = 1,6 \text{ mol} \quad n(\text{O}_2)_{\text{eq}} = 2,4 \text{ mol}$$

$$1/2 \text{ O}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l}) \quad n(\text{H}_2) = 2 \cdot n(\text{O}_2) = 2 \cdot 2,4 = 4,8 \text{ mol}$$

$$\Delta H_r^\circ = n(\text{H}_2) \cdot \Delta H_c^\circ(\text{H}_2) = 4,8 \cdot (-241,6) = -1160 \text{ kJ}$$

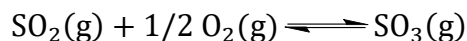
The amount of dioxygen present is sufficient.

16.2.

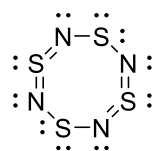
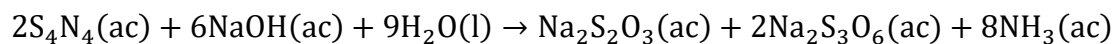
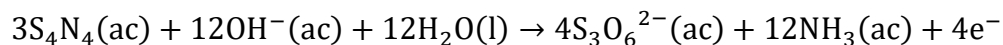
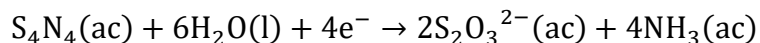
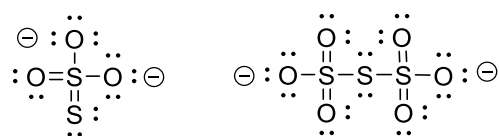
$$K_c = \frac{c(\text{SO}_3)^2}{c(\text{SO}_2)^2 \cdot c(\text{O}_2)} = \frac{4}{2,4} = 1,67$$

16.3.

The reported constant corresponds to the equilibrium

**16.4.**

A possible Lewis formula, disregarding the three-dimensional structure, is

**16.5.****16.6.**

Problem 17

17.1.

$$\alpha_2 = \frac{K_1 K_2}{[\text{H}^+]^2 + K_1 [\text{H}^+] + K_1 K_2} = 1,3 \cdot 10^{-16}$$

$$[\text{S}^{2-}] = \alpha_2 \cdot c(\text{S}^{2-}) = 1,3 \cdot 10^{-16} \cdot 0,1 = 1,3 \cdot 10^{-17} \text{ mol} \cdot \text{L}^{-1}$$

17.2.

$$\text{MeS: } [\text{Me}^{2+}][\text{S}^{2-}] = 0,01 \cdot 1,3 \cdot 10^{-17} = 1,3 \cdot 10^{-19} \text{ mol} \cdot \text{L}^{-1}$$

CoS and Ag₂S precipitate.

17.3.

$$[\text{Pb}^{2+}]_i = \sqrt{K_{ps}(\text{PbSO}_4)} = \sqrt{1,6 \cdot 10^{-8}} = 1,26 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{Pb}^{2+}]_f = \frac{K_{ps}(\text{PbS})}{[\text{S}^{2-}]} = \frac{2,5 \cdot 10^{-27}}{1,0 \cdot 10^{-17}} = 2,5 \cdot 10^{-10} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{Pb}^{2+}]_i - [\text{Pb}^{2+}]_f = 1,26 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1} \text{ which precipitate}$$

$$M(\text{PbS}) = 239,3 \text{ g} \cdot \text{mol}^{-1}$$

$$1,26 \cdot 10^{-4} \cdot 1,00 \cdot 239,3 = 0,030 \text{ g}$$

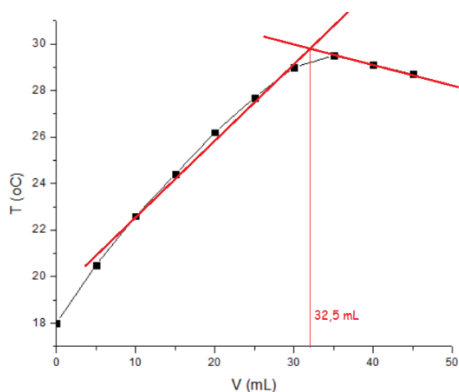
Problem 18**18.1.**

$$[\text{Ca}^{2+}] = \sqrt{K_{ps}(\text{CaCO}_3)} = \sqrt{1,2 \cdot 10^{-8}} = 1,1 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

18.2.

$$[\text{Ca}^{2+}] = [\text{CO}_3^{2-}] + [\text{C}_2\text{O}_4^{2-}] = \frac{K_{ps}(\text{CaCO}_3)}{[\text{Ca}^{2+}]} + \frac{K_{ps}(\text{CaC}_2\text{O}_4)}{[\text{Ca}^{2+}]}$$

$$[\text{Ca}^{2+}] = 1,2 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

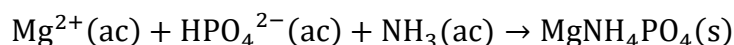
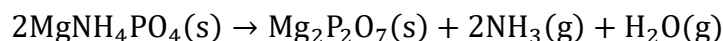
18.3.

$$n(\text{KOH}) = n(\text{HNO}_3) = c(\text{HNO}_3) \cdot V(\text{HNO}_3) = 2,00 \cdot 0,0325 = 0,065 \text{ mol}$$

$$m(\text{KOH}) = n(\text{KOH}) \cdot M(\text{KOH}) = 0,065 \cdot 56,11 = 3,65 \text{ g}$$

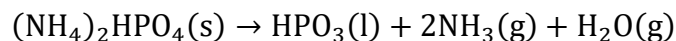
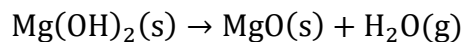
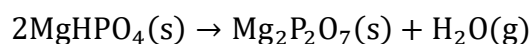
18.4.

The increase in temperature is not linear because the volume of the solution where neutralization occurs is continuously increasing.

Problem 19**19.1.****19.2.****19.3.**

$$M(\text{MgO}) = 40,31 \text{ g} \cdot \text{mol}^{-1} \quad M(\text{Mg}_2\text{P}_2\text{O}_7) = 222,56 \text{ g} \cdot \text{mol}^{-1}$$

$$\omega(\text{MgO}) = \frac{2 \cdot 40,31}{222,56} \cdot \frac{0,1532}{1,8005} \cdot 100\% = 3,08\%$$

19.4.**19.5.**

The error is positive if the percentage of magnesium in the product is lower than in $\text{Mg}_2\text{P}_2\text{O}_7$, negative if it is higher, and zero if it is the same value or if the impurities completely volatilize.

Impurity	MgHPO_4	$\text{Mg}(\text{NH}_4)_2(\text{PO}_4)_2$	$\text{Mg}_3(\text{PO}_4)_2$	$\text{Mg}(\text{OH})_2$	$(\text{NH}_4)_2\text{HPO}_4$	NH_4Cl
Error	0	+	-	-	+	0

19.6.

$$[\text{Mg}^{2+}] = \frac{0,10}{0,200 \cdot 24,31} \approx 2,1 \cdot 10^{-2} \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{OH}^-] = \sqrt{\frac{K_{\text{ps}}(\text{Mg}(\text{OH})_2)}{[\text{Mg}^{2+}]}} = \sqrt{\frac{6,0 \cdot 10^{-10}}{2,1 \cdot 10^{-2}}} = 1,7 \cdot 10^{-4} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pH} = 14,0 - \log(1,7 \cdot 10^{-4}) = 10,2$$

19.7.

$$[\text{H}^+] = 10^{-6,48} = 3,31 \cdot 10^{-7} \text{ mol} \cdot \text{L}^{-1}$$

$$\alpha_3 = \frac{K_1 K_2 K_3}{[\text{H}^+]^3 + K_1 [\text{H}^+]^2 + K_1 K_2 [\text{H}^+] + K_1 K_2 K_3} = 2,4 \cdot 10^{-7}$$

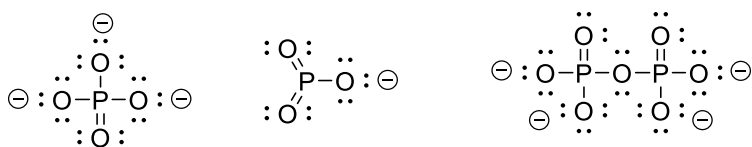
$$[\text{PO}_4^{3-}] = \alpha_3 \cdot c(\text{PO}_4^{3-}) = 2,4 \cdot 10^{-7} \cdot 0,010 = 2,4 \cdot 10^{-9} \text{ mol} \cdot \text{L}^{-1}$$

$$\text{pK}_a(\text{NH}_3) = \text{pK}_w - \text{pK}_b(\text{NH}_3) = 14,0 - \log(1,8 \cdot 10^{-5}) = 9,25$$

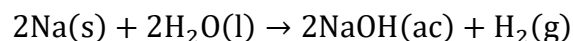
$$\text{pH} = 6,48 \ll \text{pK}_a(\text{NH}_3) = 9,25 \rightarrow [\text{NH}_4^+] \approx c(\text{NH}_4^+) = 0,010 \text{ mol} \cdot \text{L}^{-1}$$

$$[\text{Mg}^{2+}] = c(\text{MgCl}_2) = 0,010 \text{ mol} \cdot \text{L}^{-1}$$

$$K_{\text{ps}} = [\text{Mg}^{2+}][\text{NH}_4^+][\text{PO}_4^{3-}] = 0,010 \cdot 0,010 \cdot 2,4 \cdot 10^{-9} = 2,4 \cdot 10^{-13}$$

19.8.**Problem 20****20.1.**

A redox reaction is a chemical reaction in which one species loses electrons (oxidizes), and another species gains electrons (reduces). For example, in the reaction between sodium and water, metallic sodium is oxidized to Na^+ , while water is reduced to H_2 .

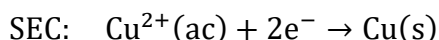
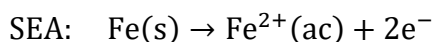


20.2.

Redox reactions: ii, iii, viii.

20.3.

The term battery or secondary cell is generally applied to semi-reversible electrochemical devices, accumulators of electrical energy, which can be recharged and only generate energy if they have been previously supplied with electricity during a charging process.

20.4.**20.5.**

It is necessary that an electrical conductor connects both electrodes and that there is a saline bridge between the solutions so that the circuit is closed.

20.6.

$$\Delta E^{\circ} = E^{\circ}(\text{cátodo}) - E^{\circ}(\text{ánodo}) = 0,34 - (-0,41) = 0,75 \text{ V}$$

20.7.

i) Hydrogen is formed and zinc is dissolved. ii) Nothing happens. iii) Iodine is formed.

20.8.

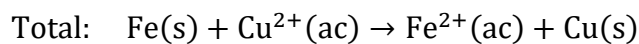
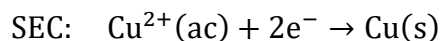
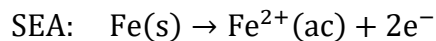
Yes. In this process, the oxidation of sulfide ions occurs, which facilitates the dissolution process. The metals remain in the form of acetates or nitrates, which are soluble salts.

20.9.

When the mixture $\text{CH}_3\text{COOH}/\text{H}_2\text{O}_2$ is employed, excess oxidizing agent can be removed by simply heating as hydrogen peroxide decomposes into non-toxic dioxygen and water.

Problem 21

21.1.



20.2.

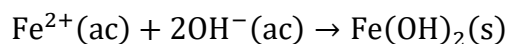
$$0,830 = \Delta E^\circ - \frac{R \cdot T}{n \cdot F} \ln \left(\frac{\text{Fe}^{2+}}{\text{Cu}^{2+}} \right)$$

$$\Delta E^\circ = 0,830 + \frac{8,314 \cdot 298}{2 \cdot 96485} \ln \left(\frac{0,0100}{0,500} \right) = 0,780 \text{ V}$$

$$\Delta E^\circ = E^\circ(\text{Cu}^{2+}/\text{Cu}) - E^\circ(\text{Fe}^{2+}/\text{Fe}) \rightarrow E^\circ(\text{Fe}^{2+}(\text{ac})/\text{Fe(s)}) = 0,340 - 0,780 = -0,440 \text{ V}$$

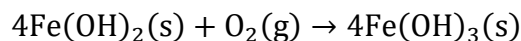
20.3.

The addition of the caustic soda solution increases the concentration of hydroxide ions. Iron(II) hydroxide is sparingly soluble and begins to form as a white precipitate.



20.4.

If there is air in the system, the iron(II) hydroxide will oxidize immediately.



20.5.

$$[\text{OH}^-] = 10^{-12,0} = 1,00 \cdot 10^{-2} \text{ mol} \cdot \text{L}^{-1}$$

$$E(\text{cathode}) = 0,340 - \frac{8,314 \cdot 298}{2 \cdot F} \ln \left(\frac{1}{0,5} \right) = 0,331 \text{ V}$$

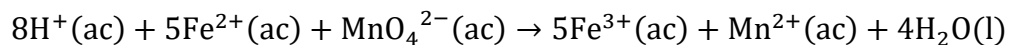
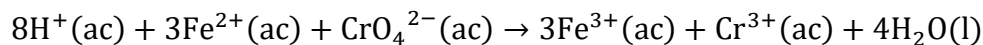
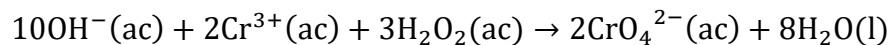
$$E(\text{anode}) = E(\text{cathode}) - \Delta E = 0,331 - 1,09 = -0,759 \text{ V}$$

$$-0,759 = -0,440 - \frac{8,314 \cdot 298}{2 \cdot F} \ln \left(\frac{1}{[\text{Fe}^{2+}]} \right) \rightarrow [\text{Fe}^{2+}] = 1,58 \cdot 10^{-11} \text{ mol} \cdot \text{L}^{-1}$$

$$K_{\text{ps}} = [\text{Fe}^{2+}][\text{OH}^-]^2 = 1,58 \cdot 10^{-11} \cdot 0,010^2 = 1,58 \cdot 10^{-15}$$

Problem 22

22.1.



Chromium content:

$$17,20 \cdot 0,020 = 0,344 \text{ mmol KMnO}_4$$

$$5 \cdot 0,344 = 1,72 \text{ mmol Fe}^{2+}$$

$$\text{reacted: } 25 \cdot 0,1 - 1,72 = 0,78 \text{ mmol Fe}^{2+}$$

to which is corresponding: $0,78/3 = 0,26 \text{ mmol Cr}$ in 150 mg of the alloy

$$m(\text{Cr}) = n(\text{Cr}) \cdot A_r(\text{Cr}) = 2,6 \cdot 52,00 = 135,2 \text{ g en 1500 mg de la aleación}$$

$$\% \text{Cr} = 9,0$$

Cu and Ag content:

$$Q(\text{total}) = 40,575 \text{ mF para 1500 mg} \quad (1087,4 \text{ mAh})$$

$$Q(\text{Cr}) = 2,6 \cdot 3 = 7,8 \text{ mF} \quad (209 \text{ mAh})$$

$$Q(\text{Cu} + \text{Ag}) = 40,575 - 7,8 = 32,775 \text{ mF} \quad (878,4 \text{ mAh})$$

$$m(\text{Cu} + \text{Ag}) = m(\text{alloy}) - m(\text{Cr}) = 1500 - 135,2 = 1364,8 \text{ mg}$$

If x is the mass of Cu present in the alloy in mg, the amount of electricity needed for the deposition in mF is

$$32,775 = \frac{2 \cdot x}{63,55} + \frac{1364,8 - x}{107,87} \rightarrow x = 906,26$$

$$m(\text{Cu}) = 906,26 \text{ mg in 1500 g of alloy}$$

$$m(\text{Ag}) = 458,54 \text{ mg in 1500 g of alloy}$$

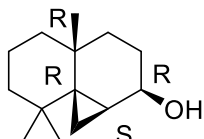
$$\% \text{Cu} = 60,4 \quad \% \text{Ag} = 30,6$$

Problem 23

23.1.

4 estereoisomer

23.2.



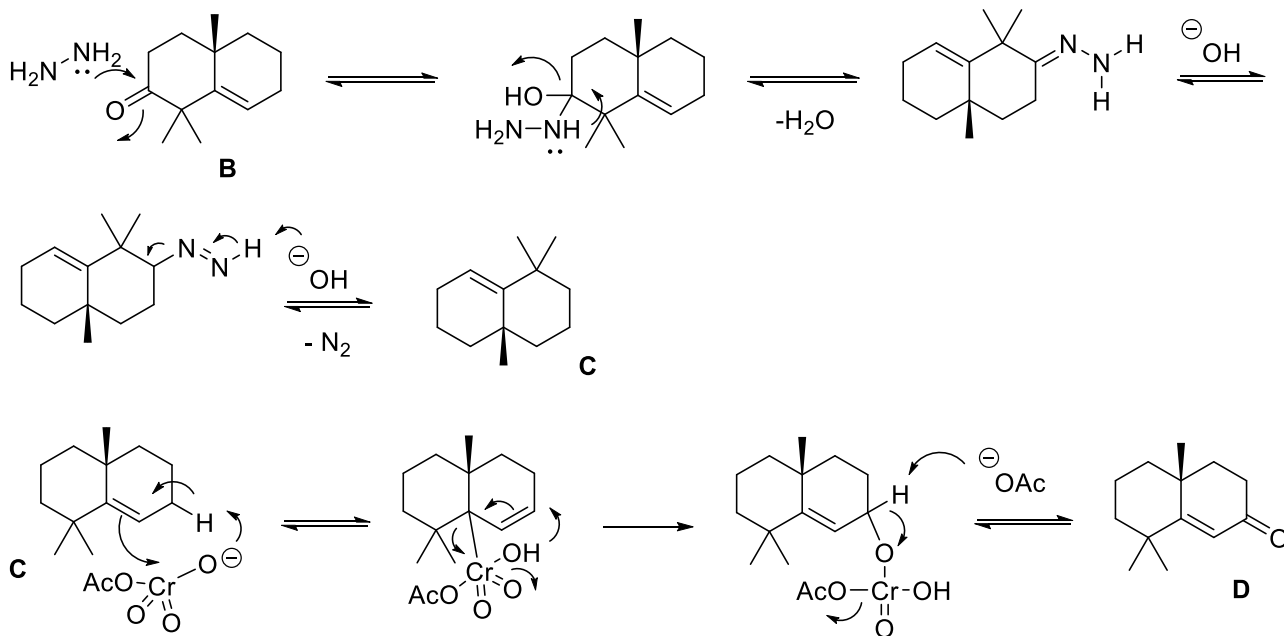
23.3.

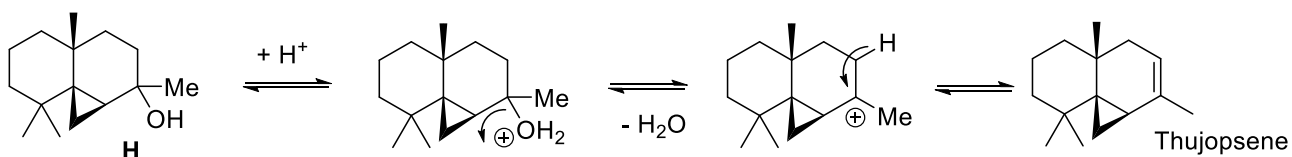
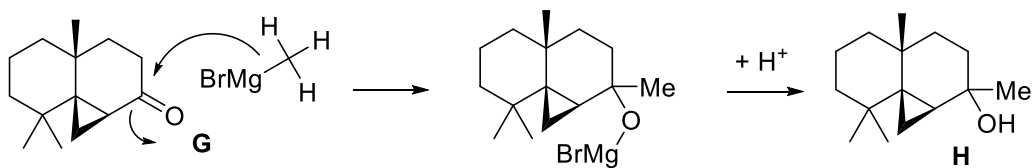
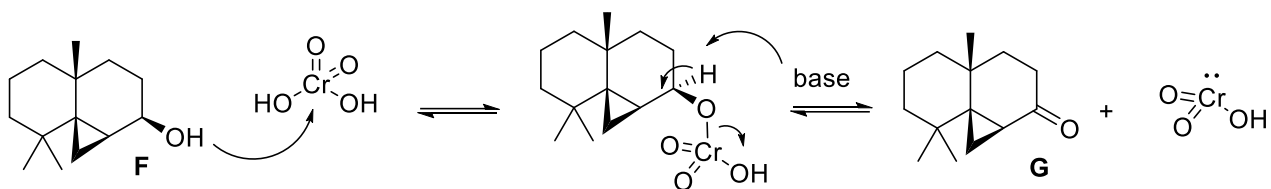
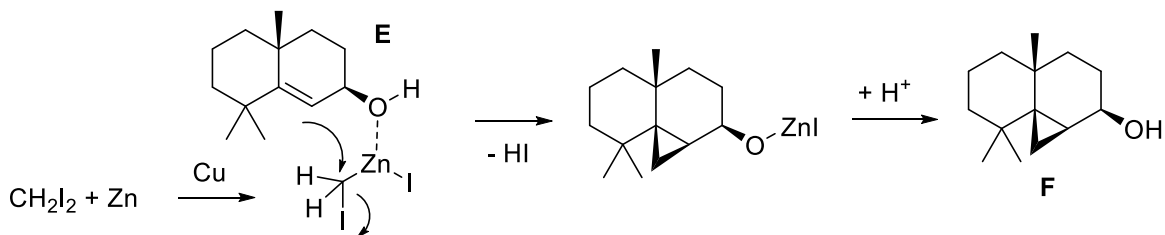
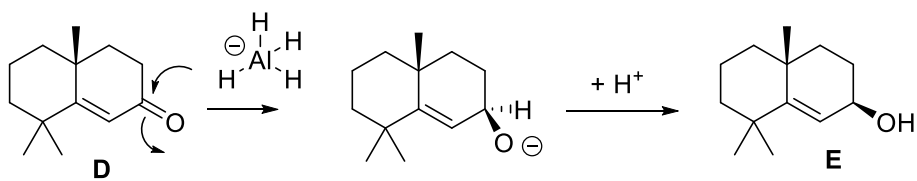
The hydride attack on the carbonyl occurs through the less hindered face of the cyclic compound **D**, which is opposite the methyl group.

23.4.

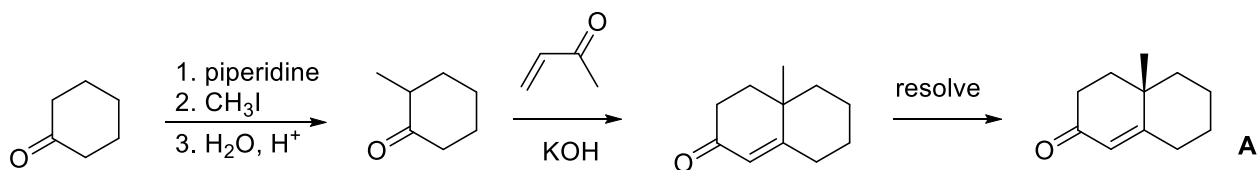
The Simmons-Smith reaction occurs syn with respect to the hydroxyl group. The oxygen atom interacts with the Zn atom of the carbenoid IZnCH_2I and stabilizes the transition state.

23.5.



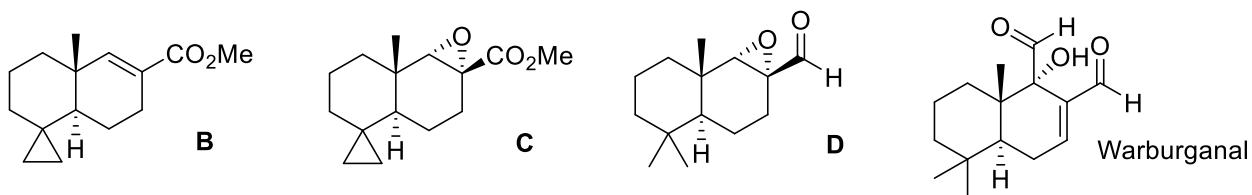


23.6.

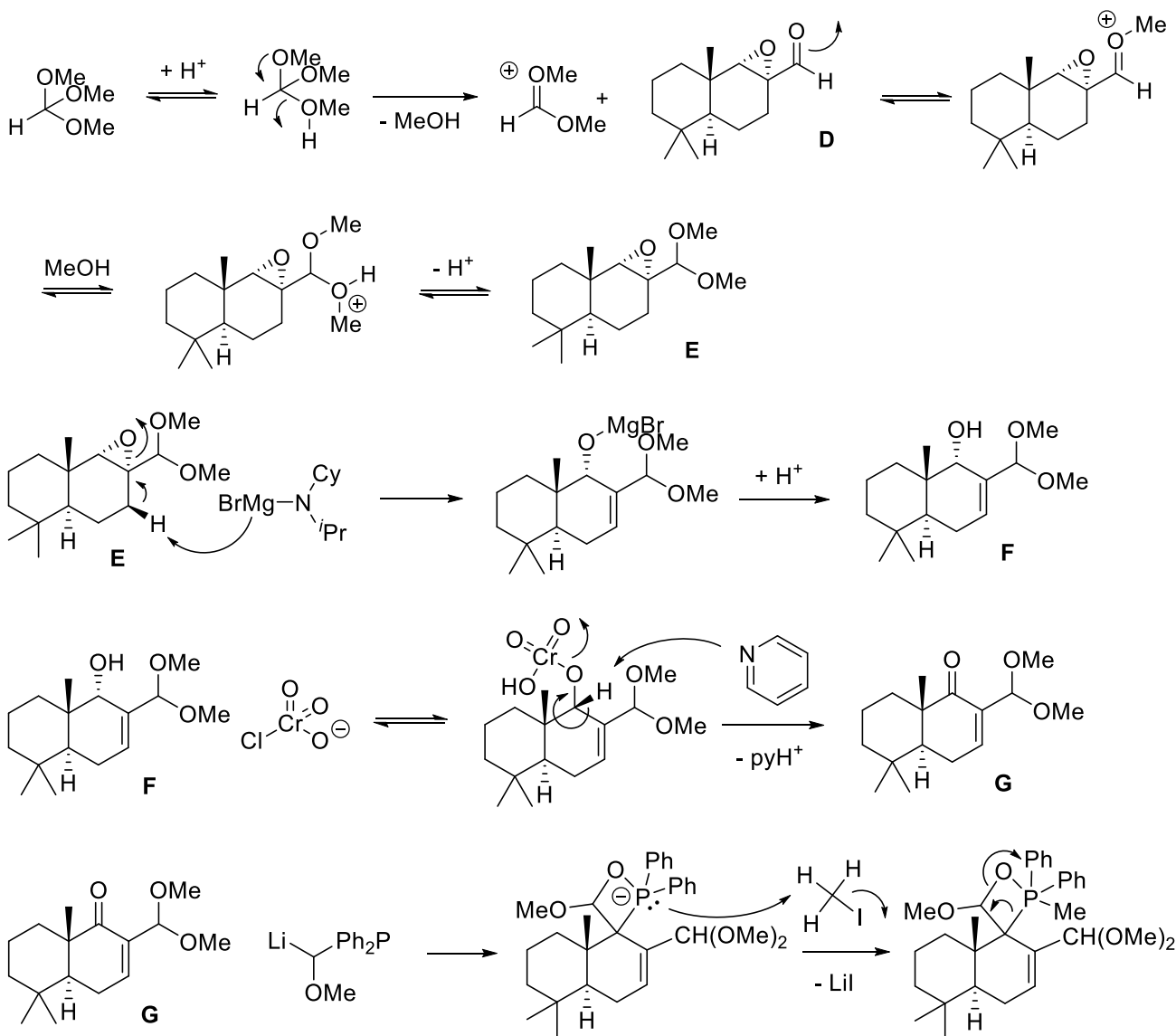


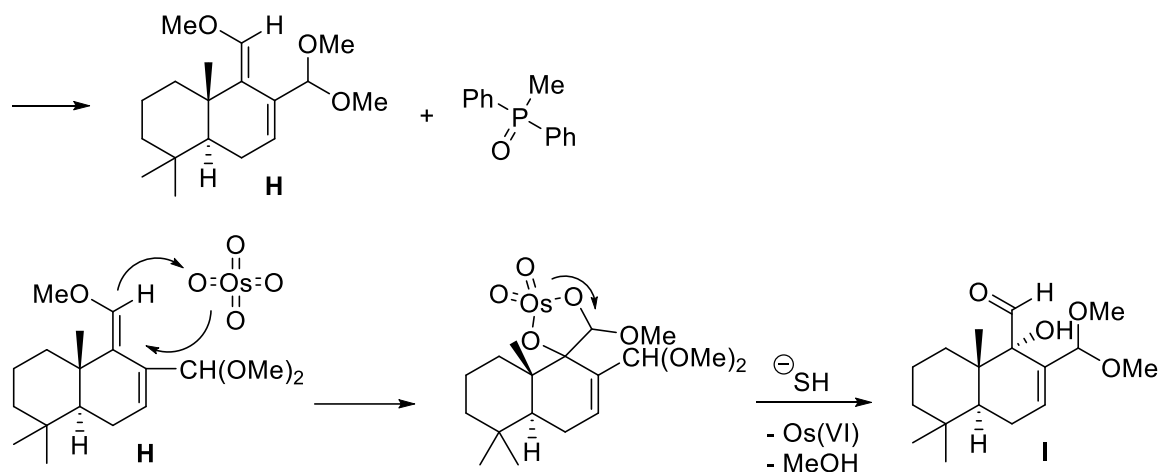
Problem 24

24.1.



24.2.





24.3.

The cyclopropane ring is highly stressed, so the corresponding C-C bonds weaken.

24.4.

Alkenes with electron-donor groups (alkyl, alkoxide) react rapidly with electrophiles, while alkenes with electro-acceptor groups (ester) react more slowly.

24.5.

2,6-dinitroperoxybenzoic acid

Problem 25

25.1.

Diels-Alder reaction. This reaction is stereospecific (the addition is *syn*), stereoselective (the *endo* product is thermodynamically favored), and regioselective (depends on the substitution pattern of the diene and dienophile).

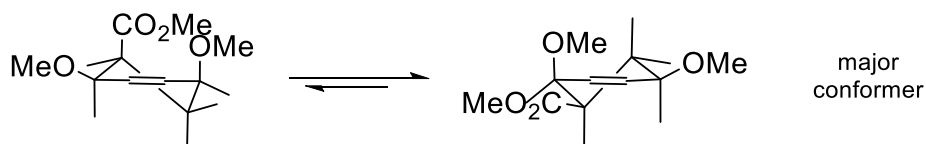
25.2.

The starting materials are achiral molecules; therefore, the product is also achiral, in this case, a racemic mixture.

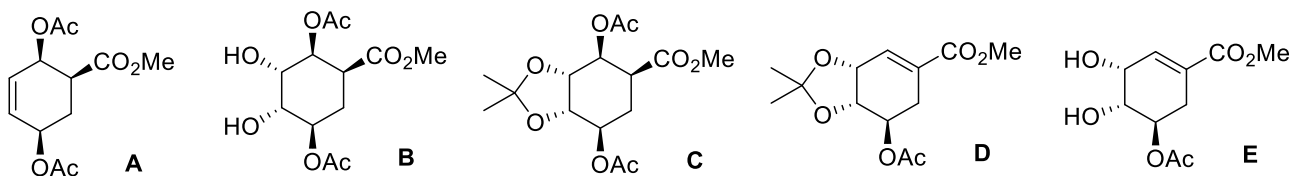
25.3.

(3R,4S,5R)-3,4,5-trihydroxycyclohex-1-ene-1-carboxylic acid

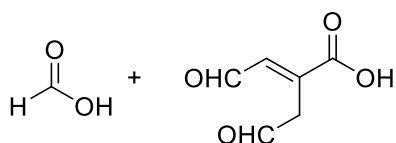
25.4.



25.5.



25.6.

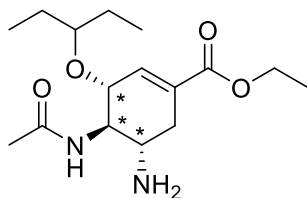


Problem 26

26.1.

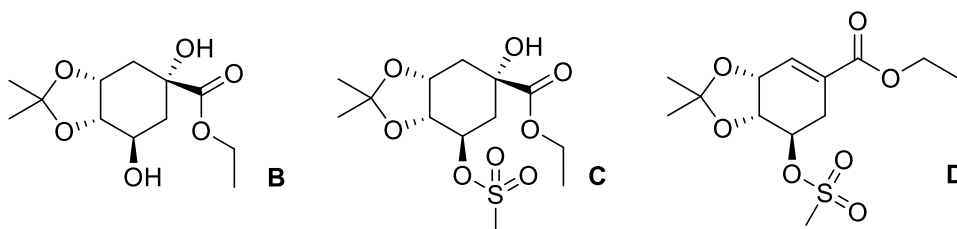
The nitrogen atom from the amine group.

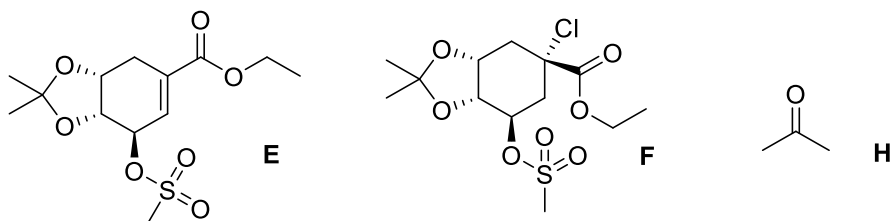
26.2.



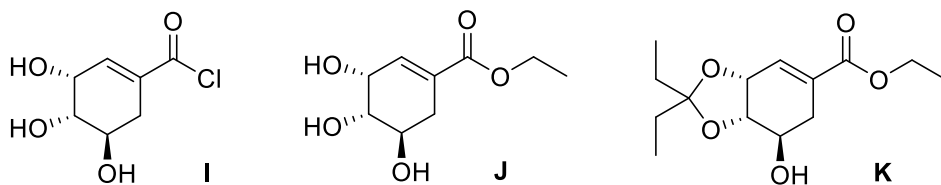
Total number of oseltamivir stereoisomers is $2^3 = 8$.

26.3.





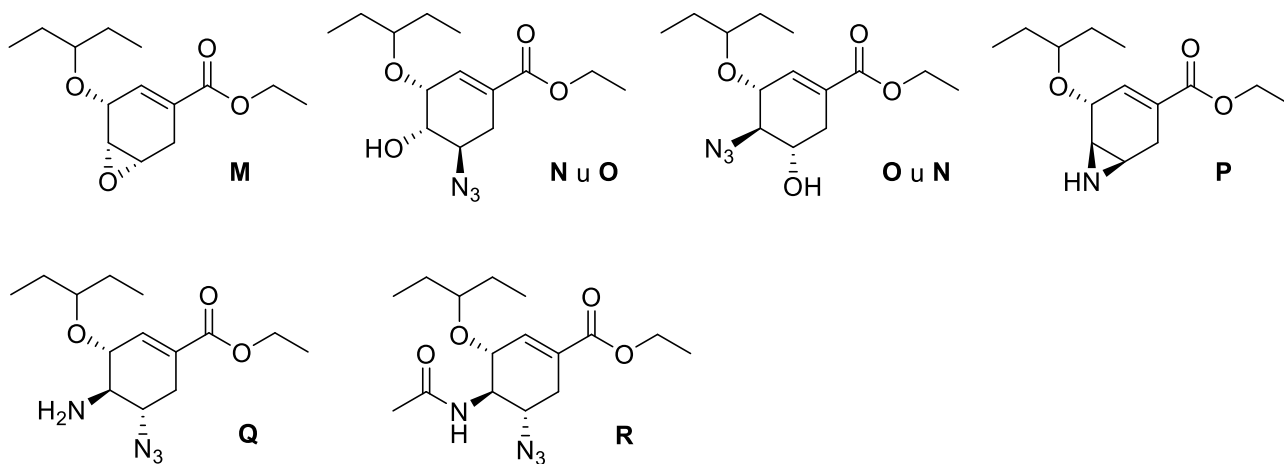
26.4.



26.5.

The hydroxyl group proton.

26.6.

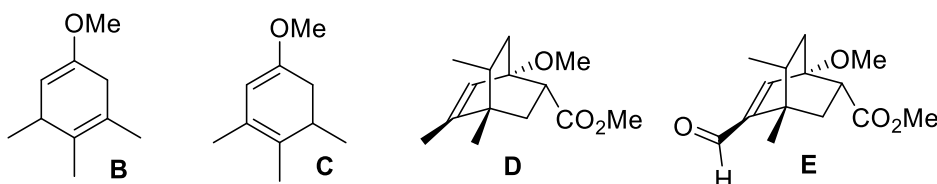


Problem 27

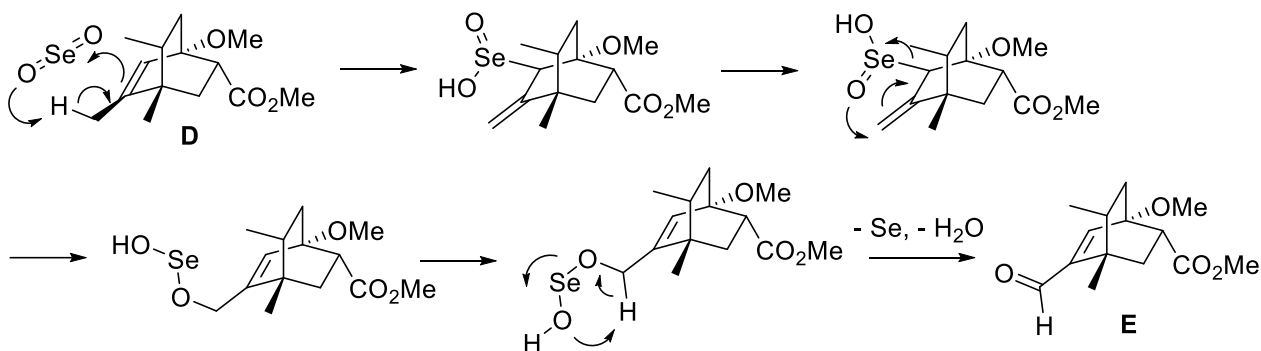
27.1.

Birch reduction. Solutions of Na, Li or K in liquid ammonia are used in the presence of an alcohol (ethanol, tert-butanol). On some occasions, an amine (propylamine, ethylenediamine) is added to the reaction mixture or other solvents (THF) are used, or an amine. It allows the reduction of substituted benzenes (with electron-donor and electro-acceptor groups), polycyclic aromatic hydrocarbons (naphthalene, anthracene, phenanthrene), pyridines and Michael acceptors (α,β -unsaturated ketones). It is a regioselective reaction.

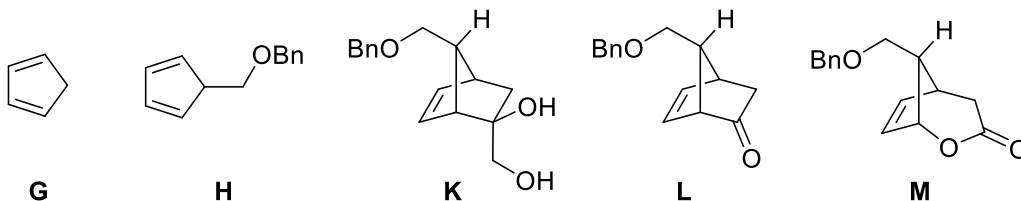
27.2.



27.3.

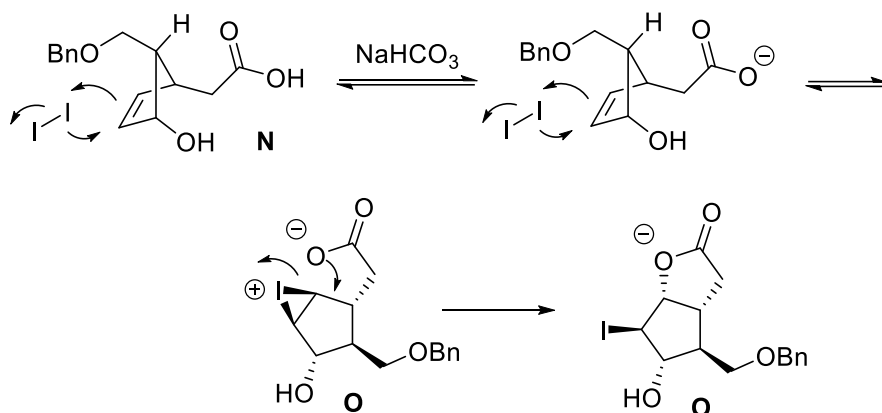


27.4.



27.5.

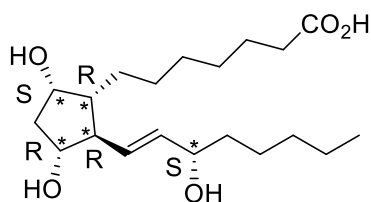
Compound **G** (cyclopentadiene) is slightly acidic ($\text{pK}_\text{a} = 16.0$) because its aromatic conjugate base is stabilized.



27.11.

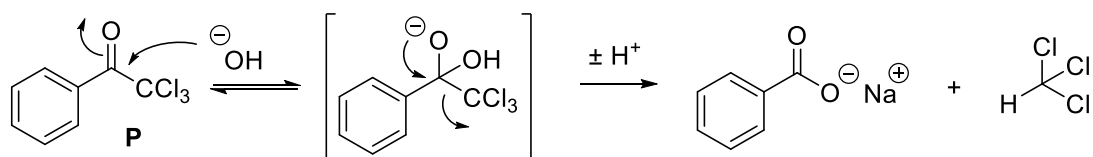
$2^4 = 32$ stereoisomers are possible. Two stereocenters form a cycle and are fixed, so they count as one.

27.12.



27.13.

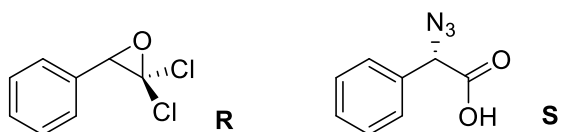
Haloform reaction.



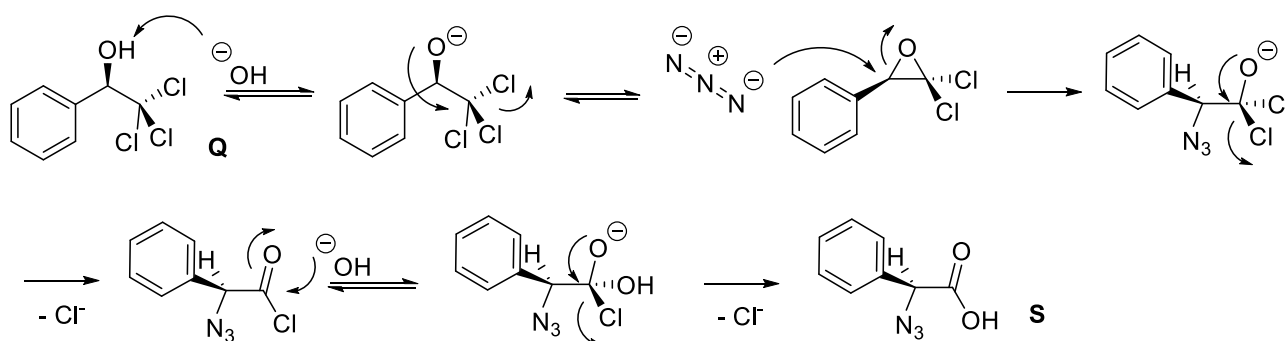
27.14.

The reduction from **P** to **Q** (known as the Corey-Bakshi-Shibata reaction or simply CBS reduction) is stereoselective.

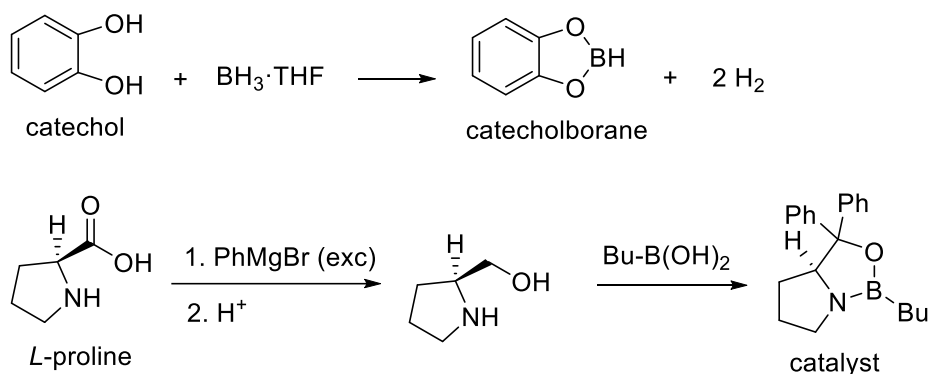
27.15.



27.16.



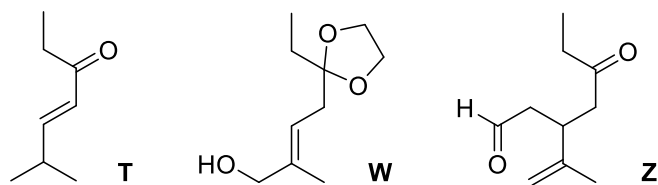
27.17.



27.18.

The reduction from **P** to **Q** (known as the Corey-Bakshi-Shibata reaction or simply CBS reduction) is stereoselective. The catalyst functions as a chiral catalyst, making the reduction stereoselective.

27.19.



27.20.

Aldolic condensation.

27.21.

The Hg(II) ion is a Lewis acid. It interacts with vinyl ether through the alkene, activating it and allowing the formation of compound **X**.

27.22.

It is a rearrangement, rearrangement, or rearrangement reaction.

Annex 1. Results of the National Chemistry Contest 2019²⁴

Results per provinces

Province	G	S	B	Total	PSN	Points	Place
CM	4	5	1	10	9	54	1
LT	1	5	4	10	6	36	2
SS	2	4	3	9	5	35	3
LH	2	3	4	9	4	31	4
MT	2	2	1	5	3	23	5
VC	0	3	2	5	2	15	6
SC	0	2	3	5	1	11	7
AR	0	1	8	9	0	11	7
PR	0	1	3	4	0	6	9
GT	0	0	5	5	0	5	10
GR	0	0	2	2	0	2	11
MY	0	0	1	1	0	1	12
IJ	0	0	1	1	0	1	12
CF	0	0	0	0	0	0	14
CA	0	0	0	0	0	0	14
HO	0	0	0	0	0	0	14
UH	0	0	0	0	0	0	14
UVC	0	0	0	0	0	0	14
UO	0	0	0	0	0	0	14

Qualification Committee

MSc. Luis Enrique Guerra Castaño	LH	IPVCE Vladimir Ilich Lenin
MSc. Rolando Alfonso Valdés	VC	IPVCE Ernesto Che Guevara
MSc. Julio Rodríguez Benítez	CM	IPVCE Máximo Gómez Baéz
MSc. Agustín Plasencia Calero	SS	IPVCE Eusebio Olivera Labra
MSc. Juan Francisco Olivera	PR	IPVCE Federico Engels
MSc. Asterio Marrero Machado	GR	IPVCE Silberto Álvarez Aroche
MSc. Rafael León Pérez	LH	IPVCE Vladimir Ilich Lenin
MSc. José María Fernández García	AT	Universidad Tecnológica de La Habana
MSc. Tomas Sánchez Herrera	GT	IPVCE José Maceo Grajales

Author of the problems

MSc. Gerardo M. Ojeda Carralero

Revision

MSc. Luis Enrique Guerra Castaño

²⁴ Symbology:

G: gold medal (5 points)

S: silver medal (3 points)

B: bronze medal (1 point)

PSN: National Preselection (2 points)

Tenth grade

Yonathan González García	MT	IPVCE Carlos Marx Pressburg	84,00	G
Anthony Cruz García	LH	IPVCE Vladimir Ilich Lenin	81,50	G
Nicolás Antonio García Álvarez	CM	IPVCE Máximo Gómez Báez	78,50	G
Samira de los Á. González Castelló	LT	IPVCE Luis Urquiza Jorge	75,00	G
Luis Antonio Lorenzo Carballo	SS	IPVCE Eusebio Olivera Rodríguez	73,00	G
Laura Medina Ramírez Medina	LT	IPVCE Luis Urquiza Jorge	67,00	S
Samira Madai Alfonso Rosa	CM	IPVCE Máximo Gómez Báez	62,50	S
José Alberto Almeida Cbrales	CM	IPVCE Máximo Gómez Báez	61,50	S
Francisco Foroni Valdivia	SS	IPVCE Eusebio Olivera Rodríguez	56,00	S
Álvaro Laterrade Pizarrosa	SC	IPU Rafael María de Mendive	51,00	S
Yan Carlos Gutiérrez	AR	IPVCE Mártires de Humboldt 7	49,50	S
Andy A. Aribé-Andudi García	MT	IPVCE Carlos Marx Pressburg	49,00	S
Daniela de la Nuez Pérez	LH	IPVCE Vladimir Ilich Lenin	48,50	S
Elisa Rodríguez Ultra	LT	IPVCE Luis Urquiza Jorge	48,50	S
Francisco M. Moreno Perdomo	VC	IPVCE Ernesto Che Guevara	47,50	S
Alejandra Gabriela Arenal García	CM	IPVCE Máximo Gómez Báez	47,00	S
Maikol Fiallo Sánchez	AR	IPVCE Mártires de Humboldt 7	44,50	B
César Alejandro Calderón Pino	SS	IPVCE Eusebio Olivera Rodríguez	44,00	B
Sheila Ortega Ruz	PR	IPVCE Federico Engels	43,50	B
José David Valles Desten	GT	IPVCE José Maceo Grajales	43,00	B
Roger Alejandro Leyva Poblador	SC	IPU Rafael María de Mendive	43,00	B
Antonio Sánchez Días	AR	IPVCE Mártires de Humboldt 7	41,50	B
Rosailis Matos Claro	GT	IPVCE José Maceo Grajales	41,50	B
Lauren Heather Anslon Rondón	LT	IPVCE Luis Urquiza Jorge	41,50	B
Leyvis Laura Serrano Jaime	PR	IPVCE Federico Engels	41,50	B
Ana Elisa Però Machirán	GT	IPVCE José Maceo Grajales	39,50	B
Eulis Gabriel Garzón Giraudi	SC	IPU Rafael María de Mendive	38,50	B
Víctor Manuel Ortiz Oliva	SS	IPVCE Eusebio Olivera Rodríguez	38,50	B
José Miguel Millán Peña	IJ	IPU América Lavadí Arce	37,00	B
Alejandro Valiente López	AR	IPVCE Mártires de Humboldt 7	36,00	B
Roxana Cárdenas Meneses	VC	IPVCE Ernesto Che Guevara	36,00	B
Alejandro Tomás Muñoz Ramírez	SS	IPVCE Eusebio Olivera Rodríguez	35,00	B
Patricia Sharon Mackey Moreira	LH	IPVCE Vladimir Ilich Lenin	34,50	B

Eleventh grade

Alessia Pérez Daniel	MT	IPVCE Carlos Marx Pressburg	87,00	G
Sheila María Peña González	CM	IPVCE Máximo Gómez Báez	86,00	G
Héctor David Couso Hernández	SS	IPVCE Eusebio Olivera Rodríguez	83,75	G
Fabio Darío Quevedo Sedeño	CM	IPVCE Máximo Gómez Báez	82,00	G
Sahiary Martínez Paz	CM	IPVCE Máximo Gómez Báez	69,75	S
Glenda Bauta Castañeda	LH	IPVCE Vladimir Ilich Lenin	66,25	S
Mariana Estrella Gamboa Céspedes	LT	IPVCE Luis Urquiza Jorge	65,00	S
Yanitza Céspedes Hechavarria	LT	IPVCE Luis Urquiza Jorge	59,50	S
Joan Ernesto Méndez Batista	LH	IPVCE Vladimir Ilich Lenin	59,00	S
Frank Aguilera Brito	VC	IPU Chichí Padrón	57,25	S
Christian Saavedra Díaz	SS	IPVCE Eusebio Olivera Rodríguez	56,50	S
Jorge A. Aragón Infante	PR	IPVCE Federico Engels	53,50	S
Laura Rodríguez Martínez	SC	IPU Rafael María de Mendive	51,75	S
Isebella Martínez Núñez	LT	IPVCE Luis Urquiza Jorge	50,75	B
Ricardo Alejandro Duarte Grinán	GT	IPVCE José Maceo Grajales	50,00	B
Sergio David Roche Díaz	LH	IPVCE Vladimir Ilich Lenin	49,00	B
Melani Casanueva Rivero	AR	IPVCE Mártires de Humboldt 7	46,00	B
Camila Viviana Rosales Olivera	GR	IPVCE Silberto Álvarez Aroche	43,50	B
Huber Cesar Cabrera Betancourt	MT	IPVCE Carlos Marx Pressburg	42,00	B
Magdiel Enrique Martínez Batista	GR	IPVCE Silberto Álvarez Aroche	41,50	B
Emy L. Hernández Montes de Oca	VC	IPVCE Ernesto Che Guevara	41,50	B
David Pérez León	AR	IPVCE Mártires de Humboldt 7	40,75	B
Greidis León Elizástigui	GT	IPVCE José Maceo Grajales	39,50	B
Lizt Adriana Pérez Díaz	PR	IPVCE Federico Engels	39,00	B
Frank Ernesto Rodríguez Carrera	MY	EMCC Mayabeque	38,00	B

Twelfth grade

Samuel Baéz Lao	LH	IPVCE Vladimir Ilich Lenin	94,00	G
Rey Michel Rodríguez Navarro	CM	IPVCE Máximo Gómez Báez	93,50	G
Osniel Alejandro García Orihuela	MT	IPVCE Carlos Marx Pressburg	82,00	S
Eduardo Adiel Landrove Escalona	LT	IPVCE Luis Urquiza Jorge	80,25	S
Rafael Alejandro Perurena García	SS	IPVCE Eusebio Olivera Rodríguez	79,25	S
Jocdan Lismar López Mantecón	CM	IPVCE Máximo Gómez Báez	78,00	S
David Linares Ramírez	VC	IPVCE Ernesto Che Guevara	77,50	S
José Manuel Cabrera Pereira	SS	IPVCE Eusebio Olivera Rodríguez	72,25	S
Marco Antonio Fernández Roquero	CM	IPVCE Máximo Gómez Báez	69,75	B
Eduardo Raúl González García	LT	IPVCE Luis Urquiza Jorge	68,50	B
Ernesto Alejandro Fajardo Díaz	AR	Colegio Universitario	67,50	B
Laura Maday De La Paz Cordero	LT	IPVCE Luis Urquiza Jorge	65,50	B
José Angel Delgado Morejón	LH	IPVCE Vladimir Ilich Lenin	63,75	B
Camilo Más Jacobs	LH	IPVCE Vladimir Ilich Lenin	63,50	B
Laura Barreto González	AR	Mártires de Humbolt 7	57,25	B
Yasser Alejandro Hernández Ravelo	AR	Colegio Universitario	51,50	B
Beatriz Reyes Manals	SC	IPVCE Antonio Maceo Grajales	51,25	B

Annex 2. Results in the international chemistry olympiads 2019

51th International Chemistry Olympiad (ICHO)

Celebrated on July 22 to 29, in Paris, France. The examinations took place in high school Pierre-Gilles de Gennes. 80 countries from all over the world participated. The Cuban delegation was:

MSc. Rolando Alfonso Valdés (Mentor)
IPVCE “Ernesto Che Guevara”, Villa Clara

Samuel Báez Lao (Student)
IPVCE “Vladimir Ilich Lenin”, La Habana, 12th grade

Rey Michel Rodríguez Navarro (Student)
IPVCE “Máximo Gómez Báez”, Camagüey, 12th grade

XXIV Iberoamerican Chemistry Olympiad (OIBQ)

Celebrated on September 7 to 15, in Porto, Portugal. The examinations took part in the University of Porto. 17 countries from Latin-America and the Iberic Peninsula took part. The Cuban delegation was:

MSc. Orestes Landrove Ramírez (Mentor)
IPVCE “Luis Urquiza Jorge”, Las Tunas

Sheila María Peña González (Student)
IPVCE “Máximo Gómez Báez”, Camagüey, 11th grade

Bronze Medal

Jocdan Lismar López Mantecón (Student)
IPVCE “Máximo Gómez Báez”, Camagüey, 12th grade

Bronze Medal

XIII Central-American and XI Caribbean Chemistry Olympiad (OCACQ)

Celebrated on August 4 to 10, in the City of Panama, Panama. The examinations took part in the University of Panama. 5 countries from Central-America and the Caribbean took part. The Cuban delegation was:

MSc. Julio Rodríguez Benítez (Mentor)
IPVCE “Máximo Gómez Báez”, Camagüey

Osniel Alejandro García Orihuela (Student)
IPVCE “Carlos Marx Pressburg”, Matanzas, 12th grade

Gold Medal

Rafael Alejandro Perurena García (Student)
IPVCE “Eusebio Olivera Rodríguez”, Sancti Spíritus, 12th grade

Gold Medal

Héctor David Couso Hernández (Student)
IPVCE “Eusebio Olivera Rodríguez”, Sancti Spíritus, 11th grade

Silver Medal

Eduardo Adiel Landrove Escalona (Student)
IPVCE “Luis Urquiza Jorge”, Las Tunas, 12th grade

Silver Medal



OCACQ: Rafael, Héctor, Julio, Eduardo and Osniel.



OIBQ: Jocdan, Sheila and Orestes.



ICHO: Rolando, Rey and Samuel with the appointed guide.



Some members of the National Preselection of Chemistry.



Annex 3. Organic Chemistry Syllabus for the National Contest

Fundamentals of structure, bonding and reactivity applied to organic molecules

Nomenclature of the main families of the organic molecules.

Stereochemistry. Fundamentals of molecular chirality, asymmetric center, plane and symmetry axes; atoms that can be asymmetric (C, S, N, P); chirality of a molecule, chirality in allenes and biphenyls; absolute stereochemistry and relative stereochemistry; Cahn-Ingold-Prelog notation; “saw-horse” structure; E and Z notation for alkenes; types of stereoisomers, chemical and physical properties, optical activity; racemization and racemic mixture; meso, erithro and threo structures.

Conformational analysis. Conformations of alkanes and cycloalkanes; angular tension and sterical tension; staggered, anti and gauche conformations; envelope conformation of cyclopentane; chair and boat conformations, axial and equatorial positions; Newman and saw-horse projections; cis and trans fusing in cyclic compounds; Bredt’s rule for bridged bicycles.

Intermediaries in organic reactions: carbocations, carbanions, radicals and carbenes; formation; relative stability.

Stereoelectronic effects. Resonance, conjugation and hyperconjugation; inductive effect; mesomeric effect; aromaticity and antiaromaticity, Hückel’s rules; π -excess and π -deficient rings; participation in vicinal group (anchimeric assistance).

Reactivity. Nucleophile and electrophile; leaving groups; acidity and basicity of organic molecules, influencing factors; comparison of S_N1 , S_N2 , E1 and E2, factors influencing the mechanism (electronic effects, sterical hindrance, strength of the base/nucleophile, solvent and temperature, kinetics, transpositions); quantitative calculations of the halogenation, difference between reactivity and selectivity, Hammond’s postulate; Markovnikov’s rule for the electrophilic addition of alkenes and alkynes, peroxide or Kharasch-Mayo effect in the addition of HBr to alkenes; Pearson’s theory of hard and soft acid and bases, Michael’s addition vs 1,2-addition, S_N2 and S_N2' reactions in allylic systems; causes of the different reactivity of alkenes, alkynes, dienes and aromatic derivatives; kinetic control and thermodynamic control in the reactions of conjugated dienes (1,2 and 1,4 products) and in the Diels-Alder’s reactions (*endo* and *exo* products); relative stability of σ complex in S_NAr and S_EAr reactions, activating and deactivating groups, regioselectivity of the reactions in the substituted benzene derivatives, furane, pyrrole, thiophene, pyridine, naphthalene, anthracene and phenanthrene;

regioselectivity and stereoselectivity of the chemical reactions; relative reactivity of the carbonylic compounds towards nucleophiles, influence of effects and electronics steric and the living groups.

Radical reactions

Alkane halogenations; Kolbe's synthesis of alkanes; allylic and benzylic halogenation; Hunsdiecker's reactions; alkane pyrolysis and combustion reactions.

Organometallic reagents

Oxidative insertion of Mg or Li in alkyl, aryl and vinyl halides; alkyne disproportionation; reaction with carbon dioxide, formaldehyde, aldehydes and ketones; transmetalation reactions to form other organometallic reagents (Cu, Zn, Cd); coupling reactions between the organometallic reagents and halides (Corey-House's synthesis, Wurtz's reactions, Ullmann's reactions).

Nucleophilic substitution in the saturated carbon

S_N1 and S_N2 reactions of halides, sulfonates (tosylates, mesylates, triflates) and alike; reactions with water, silver salts, thiols, alcohols, phosphines, amines, azide, cyanide, alkynes, Gilman's reagents, phthalimide (Gabriel's synthesis), phenols (Williamson's synthesis) and thiophenols, halides (Filkenstein's reactions), (thio)carboxylic acids; reaction of alcohols with hydric acids; exhaustive methylation with Me₂SO₄ and MeI; methylation of phenols and carboxylic acids with diazomethane; reaction of alcohols with phosphorous halides; deprotection of phenolic ethers with BBr₃; epoxide opening in basic and acid media with water, alcohols, amines, cyanides, hydrides, azide; Ritter's reactions; epoxide formation from halohydrins; reaction of ethers with hydrogen halides and water in acid media.

Elimination reactions

E1 and E2 reactions with different kind of bases and leaving groups; Hofmann's and Zaitzev's orientation; Hofmann's elimination; Cope's elimination; elimination in carbocations; alcohols dehydration; alkyne formation from vicinal dihalides and vinyl halide, regioselectivity; conjugate elimination (E_{1c}B) in the products of aldolic additions and alike.

Electrophilic addition to alkenes and alkynes

Reaction of alkenes with halogens and similar (IBr, ClSR), peroxides (Prilezhaev's reactions), hydrogen halide, N-halosuccinimides, sulfuric acid, boranes; Simmons-Smith's reactions; ozonolysis (oxidative, reductive and neutral), halohydrins formation and similar; oxy(alcoxy)mercuration-

demercuration; electrophilic addition to dienes at low and high temperature; alkene and alkyne hydration; acetylene polymerization.

Electrophilic addition to alkenes

Reactions of double bonds conjugated to electron-withdrawing groups (amides, esters, ketones, aldehydes, acyl chloride, nitro...) with cyan hydric acid, Grignard's reagent, Gilman's reagents, organolithiums, ammonia, amines, sodium borohydride, lithium and aluminum hydride, thiols, alcohols, phenols, thiophenols, carboxylic acid salts, phosphines, peroxides, hydrogen halides, acetylides and enolates (Michael's reactions).

Nucleophilic addition to the carbonyl group

Cyanohydrin formation; reaction of aldehydes and ketones with hydrides, organometallic reagents, water, alcohols and bisulfite.

Nucleophilic substitution at the carbonyl group with loss of the oxygen atom

Formation of linear and cyclic (a)ketals; formation of 1,3-dithians; reaction of aldehydes and ketones with amines, hydroxylamine, hydrazine, (2,4-dinitro)phenylhydrazine and semicarbazide; reaction with POCl_3 and PCl_5 ; reductive amination; Strecker's synthesis; Wittig's reaction.

Nucleophilic substitution at the carbonyl group

Relative reactivity of acid chloride, anhydrides, esters, amides and carboxylic acids; reaction with amines, thiols, alcohols, water and carboxylic acids in basic or acid media; reaction of carboxylic acids with SOCl_2 , PCl_5 and oxalyl chloride; partial and total hydrolysis of nitriles; reactions with organolithiums, Grignard's and Gilman's reagents; reduction with lithium aluminum hydride and diborane; sulfonamides and tosylates formation; addition of amines and alcohols to isocyanates; reaction of diazomethane with acid halide.

Reactions of enols, enolates and similar

Enamine alkylation and acylation; formation of enols and enolates of esters, ketones, aldehydes, acid halides, nitriles, sulfones, sulfoxides and nitro compounds; reaction of enols and enolates with alkyl halides, acyl halides, sulfonic acid esters, halogens and their equivalents (NIS), epoxides, nitrous acid; haloform reaction; substitution reaction of 1,3-dithianes; aldolic addition and condensation; Claisen's and Dieckmann's reaction; malonic synthesis, decarboxylation of malonic acid; acetoacetic synthesis, decarboxylation of α -ketoacids and α -ketoesters (Krapcho's reaction); Hell-Volhard-Zelinsky's

reaction; analog reactions to aldolic and Claisen's reactions in compounds with active hydrogens (reactions of Knoevenagel, Doebner, Perkin, nitroaldol or Henry, Reformatsky, Stobbe, Mannich, Darzens...).

Aromatic electrophilic substitution

Halogenation, nitration, sulfonation of benzene and its derivatives; Friedel-Crafts's alkylation and acylation; Kolb -Schmidt's carbonation; Gattermann and Gattermann-Koch's reactions; Reimer-Tiemann's reactions; Hoesch's reactions; chloromethylation; nitrosation.

Nucleophilic aromatic substitution

Reaction of deactivated halobenzene with amines, alcohols, thiols, water, hydrazine, cyanide, phenols, thiophenols and azide.

Cycloaddition reactions

Diels-Alder's reaction; alkene hydroxylation with OsO_4 ; ozonolysis; carbenes insertion to alkenes, Simmons-Smith's reaction.

Rearrangements and fragmentations

Pinacolinic and benzylic acid rearrangements; Baeyer-Villiger's and Dakin's reactions; migrations in carbocations; Beckman's rearrangements; Hofmann's transposition; Tischenko's reaction.

Oxidations and reductions

Reduction reactions with LiAlH_4 , NaBH_4 , H_2 /catalyst, B_2H_6 , N_2H_2 , NaBH_3CN , DIBALH, metal/ $\text{NH}_3(\text{l})$; Birch's reduction, $\text{Zn}(\text{Hg})/\text{HCl}$ (Clemmensen's reduction), Sn/HCl , NH_2NH_2 /base (Wolf-Kirchner's reaction), $\text{Na}/\text{alcohol}$; pinacols obtention.

Oxidation reactions with KMnO_4 , HIO_4 , H_2CrO_4 , PCC, PDC, H_2O_2 , MCPBA, O_3 , OsO_4 . Cannizaro's reaction; Meerwein-Ponndorf-Verley's reduction and Oppenauer's oxidation; allylic oxidations with CrO_3 and SeO_2 ; Riley's oxidation.

General mechanism of the oxidation reactions of alcohols and aldehydes.

Other reactions

Amides and oximes dehydration; addition of Grignard's reagent to nitriles; formation of diazonium salts; reaction of diazonium salts with CuX , KI , water, H_3PO_2 , anilines, phenols, malonic and acetoacetic esters; installation and removal of protecting groups (thio)aketal, THP, Bn, Me, Et, tBu,

TMS, Boc, Cbz; formation of sodium, copper and silver acetylides; Arndt-Eistert's reaction; Nef's reaction; oxidation of boranes with peroxide to form alcohols.

Identification of organic molecules

Hinsberg's test; haloform reaction; Lucas's test; reaction with Na; reaction with bromine water; Baeyer's reagent; Tollens' reagents; Fehling's reagent; Benedict's reagent; reaction with AgNO_3 ; reaction with FeCl_3 ; reaction with HCl ; reaction with NaOH ; reaction with Na_2CO_3 ; reaction with NaHCO_3 ; Jones' reagent; reaction with NaI and NaIO_3 aqueous solution; alkaline fusion with Na followed by the reaction with AgNO_3 , PbAc_2 or $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NOS}]$ and FeSO_4 .